

COMPLEX NUMBER

1. INTRODUCTION :

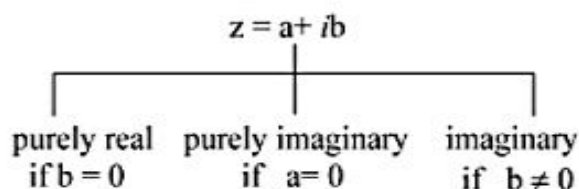
Indian mathematician Mahavira (850 A.D.) was first to mention in his work 'Ganitasara Sangraha' that as in nature of things a negative (quantity) is not a square (quantity), it has, therefore, no square root. Hence there is no real number x which satisfies the polynomial equation $x^2 + 1 = 0$.

A symbol $\sqrt{-1}$, denoted by letter i was introduced by Swiss Mathematician, Leonhard Euler (1707-1783) in 1748 to provide solutions of equation $x^2 + 1 = 0$. i was regarded as a fictitious or imaginary number which could be manipulated algebraically like an ordinary real number, except that its square was -1 . The letter i was used to denote $\sqrt{-1}$, possibly because i is the first letter of the Latin word 'imaginarium'.

To permit solutions of such polynomial equations, the set of complex numbers is introduced. We can consider a complex number as having the form $a + ib$ where a and b are real numbers. It is denoted by z i.e. $z = a + ib$. ' a ' is called as real part of z which is denoted by $\text{Re}(z)$ and ' b ' is called as imaginary part of z which is denoted by $\text{Im}(z)$.

1.1 Classification of complex number :

In fact every complex can be classified as

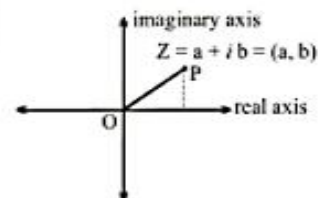


Hence, $0 + 0i$ is both a purely real as well as a purely imaginary but not imaginary.

1.2 Geometrical representation of a complex number :

Master Argand had done a systematic study on complex numbers and represented every complex number as a set of ordered pair (a, b) on a plane called complex plane / Argand plane.

All complex numbers lying on the real axis were called as purely real and those lying on the imaginary axis as purely imaginary.



1.3 Integral Powers of i :

We have $i = \sqrt{-1}$ so $i^2 = -1$, $i^3 = -i$, $i^4 = 1$
 or $i^{4n+1} = i$, $i^{4n+2} = -1$ for any $n \in \mathbb{I}$,
 $i^{4n+3} = -i$, $i^{4n} = 1$

Thus any integral power of i can be expressed as ± 1 or $\pm i$.

Illustration :

Find the value of $\frac{i^{592} + i^{590} + i^{588} + i^{586} + i^{584}}{i^{582} + i^{580} + i^{578} + i^{576} + i^{574}} + 2$.

Sol.
$$\frac{i^{592} + i^{590} + i^{588} + i^{586} + i^{584}}{i^{582} + i^{580} + i^{578} + i^{576} + i^{574}} + 2 = \frac{i^{584}}{i^{574}} + 2$$
$$= i^{10} + 2 = -1 + 2 = 1$$

Note :

- (a) The set $\mathbb{R}$ of real number is a proper subset of the Complex Numbers. Hence the complete number system is $\mathbb{N} \subset \mathbb{W} \subset \mathbb{I} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$.
- (b) Zero is purely real as well as purely imaginary but not imaginary.
- (c) $\sqrt{a}\sqrt{b} = \sqrt{ab}$ only if atleast one of a or b is non-negative.
- (d) $z_1^2 + z_2^2 = 0 \Rightarrow z_1 = 0 = z_2$ i.e., $z_1 = 1 + i$ and $z_2 = 1 - i$

2. ALGEBRA OF COMPLEX NUMBER :

2.1 Equality of complex number :

Let there be two complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$.

If $z_1 = z_2$ then $\text{Re}(z_1) = \text{Re}(z_2)$ and $\text{Im}(z_1) = \text{Im}(z_2)$.

i.e., if $x_1 + iy_1 = x_2 + iy_2$

$\Rightarrow x_1 = x_2$ and $y_1 = y_2$ simultaneously.

Illustration :

Find the real values of x and y for which the equation $(x^4 + 2xi) - (3x^2 + yi) = (3 - 5i) + (1 + 2yi)$ is satisfied.

Sol. Given equation $(x^4 + 2xi) - (3x^2 + yi) = (3 - 5i) + (1 + 2yi)$

$\Rightarrow (x^4 - 3x^2) + i(2x - 3y) = 4 - 5i$

Equating real and imaginary parts, we get

$x^4 - 3x^2 = 4 \quad \dots (i)$

and $2x - 3y = -5 \quad \dots (ii)$

From (i) and (ii), we get $x = \pm 2$ and $y = 3, \frac{1}{3}$

Note : Inequality in complex numbers are never talked. If $x_1 + iy_1 > x_2 + iy_2$ has to be meaningful $\Rightarrow y_1 = y_2 = 0$. Equalities however in complex numbers are meaningful. Two complex numbers z_1 and z_2 are said to be equal if

$\text{Re } z_1 = \text{Re } z_2$ and $\text{Im } (z_1) = \text{Im } (z_2)$ (i.e. they occupy the same position on complex plane)

2.2 Addition :

$$z_1 + z_2 = (x_1 + i y_1) + (x_2 + i y_2) = (x_1 + x_2) + i (y_1 + y_2) \in \mathbb{C}.$$

It is easy to observe that the sum of two complex numbers is a complex number whose real (imaginary) part is the sum of the real (imaginary) parts of the given numbers :

$$\operatorname{Re}(z_1 + z_2) = \operatorname{Re}(z_1) + \operatorname{Re}(z_2);$$

$$\operatorname{Im}(z_1 + z_2) = \operatorname{Im}(z_1) + \operatorname{Im}(z_2)$$

2.3 Subtraction :

$$z_1 - z_2 = (x_1 + i y_1) - (x_2 + i y_2) = (x_1 - x_2) + i (y_1 - y_2) \in \mathbb{C}.$$

That is

$$\operatorname{Re}(z_1 - z_2) = \operatorname{Re}(z_1) - \operatorname{Re}(z_2);$$

$$\operatorname{Im}(z_1 - z_2) = \operatorname{Im}(z_1) - \operatorname{Im}(z_2).$$

2.4 Multiplication :

$$z_1 \cdot z_2 = (x_1 + i y_1) (x_2 + i y_2) = (x_1 x_2 - y_1 y_2) + i (x_1 y_2 + x_2 y_1) \in \mathbb{C}.$$

In other words

$$\operatorname{Re}(z_1 z_2) = \operatorname{Re}(z_1) \cdot \operatorname{Re}(z_2) - \operatorname{Im}(z_1) \cdot \operatorname{Im}(z_2)$$

and $\operatorname{Im}(z_1 z_2) = \operatorname{Im}(z_1) \cdot \operatorname{Re}(z_2) + \operatorname{Im}(z_2) \cdot \operatorname{Re}(z_1)$

For a real number λ and a complex number $z = x + i y$.

$$\lambda \cdot z = \lambda (x + i y) = \lambda x + i \lambda y \in \mathbb{C}$$

is the product of a real number with a complex number. The following properties are obvious :

(a) $\lambda (z_1 + z_2) = \lambda z_1 + \lambda z_2$

(b) $\lambda_1 (\lambda_2 z) = (\lambda_1 \lambda_2) z;$

(c) $(\lambda_1 + \lambda_2) z = \lambda_1 z + \lambda_2 z$ for all $z, z_1, z_2 \in \mathbb{C}$ and $\lambda, \lambda_1, \lambda_2 \in \mathbb{R}.$

Actually, relations (a) and (c) are special cases of the distributive law and relation (b) comes from the associative law of multiplication for complex numbers.

2.5 Division of Complex Number :

Let $z_1 = x_1 + i y_1$ & $z_2 = x_2 + i y_2$

Then $\frac{z_1}{z_2} = \frac{x_1 + i y_1}{x_2 + i y_2} \Rightarrow \frac{(x_1 + i y_1)(x_2 - i y_2)}{(x_2 + i y_2)(x_2 - i y_2)}$

$$\Rightarrow \frac{(x_1 x_2 + y_1 y_2) + i (y_1 x_2 - x_1 y_2)}{(x_2^2 + y_2^2)} \Rightarrow \left\{ \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} \right\} + i \left\{ \frac{y_1 x_2 - x_1 y_2}{x_2^2 + y_2^2} \right\}$$

$$\frac{z_1}{z_2} \Rightarrow \operatorname{Re}\left(\frac{z_1}{z_2}\right) + i \operatorname{Im}\left(\frac{z_1}{z_2}\right)$$

2.6 Square Root of Complex Number :

Let $z = x + iy$ be the given complex number and we have to obtain its square root.

$$\text{Let } a + ib = (x + iy)^{1/2} \Rightarrow a^2 - b^2 + 2iab = x + iy$$

$$\Rightarrow x = a^2 - b^2 \text{ and } y = 2ab$$

$$\Rightarrow x^2 = (a^2 - b^2)^2 - 4a^2b^2 \Rightarrow x^2 + y^2 = (a^2 + b^2)^2$$

$$\Rightarrow a^2 + b^2 = |z| \quad \dots\dots(1) \quad \Rightarrow a^2 - b^2 = x \quad \dots\dots(2)$$

$$\Rightarrow a^2 = \frac{|z| + x}{2} \Rightarrow a = \pm \sqrt{\frac{|z| + x}{2}} \quad \Rightarrow b^2 = \frac{|z| - x}{2} \Rightarrow b = \pm \sqrt{\frac{|z| - x}{2}}$$

$$\therefore \sqrt{x + iy} = a + ib = \pm \left(\sqrt{\frac{|z| + \text{Re}(z)}{2}} + i \sqrt{\frac{|z| - \text{Re}(z)}{2}} \right)$$

Replacing i by $-i$, we get

$$\sqrt{x - iy} = \pm \left(\sqrt{\frac{|z| + \text{Re}(z)}{2}} - i \sqrt{\frac{|z| - \text{Re}(z)}{2}} \right)$$

Illustration :

Find the square root of $3 + 4i$

$$\text{Sol. Let } \sqrt{3 + 4i} = a + ib \Rightarrow 3 + 4i = a^2 - b^2 + 2iab \Rightarrow a^2 - b^2 = 3, 2ab = 4$$

$$\therefore a^2 + b^2 = \sqrt{(a^2 - b^2)^2 + 4a^2b^2} = \sqrt{9 + 16} = 5$$

$$\therefore a + ib = \pm (2 + i)$$

Alternative method :

$$\text{Using formula } \sqrt{3 + 4i} = \pm \left(\sqrt{\frac{5 + 3}{2}} + i \sqrt{\frac{5 - 3}{2}} \right) = \pm (2 + i)$$

Illustration :

If $z = x + iy$, $z^{1/3} = a - ib$ and $\frac{x}{a} - \frac{y}{b} = k(a^2 - b^2)$, then find the value of k .

$$\text{Sol. } (x + iy)^{1/3} = a - ib$$

$$\Rightarrow x + iy = (a - ib)^3 = (a^3 - 3ab^2) + i(b^3 - 3a^2b)$$

$$\Rightarrow x = a^3 - 3ab^2, y = b^3 - 3a^2b \Rightarrow \frac{x}{a} = a^2 - 3b^2 \text{ and } \frac{y}{b} = b^2 - 3a^2$$

$$\Rightarrow \frac{x}{a} - \frac{y}{b} = a^2 - 3b^2 - b^2 + 3a^2 = 4(a^2 - b^2)$$

$$\therefore k = 4.$$

Illustration :

Find the values of θ if $\frac{(3+2i\sin\theta)}{(1-2i\sin\theta)}$ is purely real or purely imaginary.

Sol. $z = \frac{3+2i\sin\theta}{1-2i\sin\theta}$

Multiplying numerator and denominator by conjugate,

$$z = \frac{(3+2i\sin\theta)(1+2i\sin\theta)}{1+4\sin^2\theta} = \frac{3-4\sin^2\theta+8i\sin\theta}{1+4\sin^2\theta}$$

Now z is purely real if $\sin\theta = 0$ or $\theta = n\pi, n \in I$. z is purely imaginary if $3-4\sin^2\theta = 0$

$$\Rightarrow \sin\theta = \pm \frac{\sqrt{3}}{2} = \pm \sin\frac{\pi}{3} \Rightarrow \theta = n\pi \pm \frac{\pi}{3}, n \in I.$$

Illustration :

Show that the polynomial $x^{4p} + x^{4q+1} + x^{4r+2} + x^{4s+3}$ is divisible by $x^3 + x^2 + x + 1$ where $p, q, r, s \in N$.

Sol. Let $f(x) = x^{4p} + x^{4q+1} + x^{4r+2} + x^{4s+3}$

$$x^3 + x^2 + x + 1 = (x^2 + 1)(x + 1) = (x + i)(x - i)(x + 1)$$

$$f(i) = i^{4p} + i^{4q+1} + i^{4r+2} + i^{4s+3} = 1 + i^1 + i^2 + i^3 = 1 + i - 1 - i = 0$$

$$f(-i) = (-i)^{4p} + (-i)^{4q+1} + (-i)^{4r+2} + (-i)^{4s+3} \\ = 1 + (-i)^1 + (-i)^2 + (-i)^3 = 1 - i - 1 + i = 0$$

$$f(-1) = (-1)^{4p} + (-1)^{4q+1} + (-1)^{4r+2} + (-1)^{4s+3} = 0$$

Thus by division theorem $f(x)$ is divisible by $x^3 + x^2 + x + 1$.

Illustration :

If the expression $(1 + ir)^3$ is of the form of $s(1 + i)$ for some real 's' where 'r' is also real and $i = \sqrt{-1}$, then the value of 'r' can be

(A) $\cot\frac{\pi}{8}$ (B) $\sec\pi$ (C) $\tan\frac{\pi}{12}$ (D) $\tan\frac{5\pi}{12}$

Sol. We have $(1 + ri)^3 = s(1 + i)$

$$1 + 3ri + 3r^2i^2 + r^3i^3 = s(1 + i)$$

$$1 - 3r^2 + i(3r - r^3) = s + si \Rightarrow 1 - 3r^2 = s = 3r - r^3$$

$$\text{Hence } 1 - 3r^2 = 3r - r^3$$

$$\Rightarrow r^3 - 3r^2 - 3r + 1 = 0 \Rightarrow (r^3 + 1) - 3r(r + 1) = 0 \Rightarrow (r + 1)(r^2 + 1 - r - 3r) = 0$$

$$\therefore r = -1 \text{ or } r^2 - 4r + 1 = 0$$

$$\Rightarrow r = \frac{4 \pm \sqrt{16-4}}{2} = \frac{4 \pm 2\sqrt{3}}{2} \Rightarrow r = 2 + \sqrt{3} \text{ or } 2 - \sqrt{3} \Rightarrow B, C, D$$

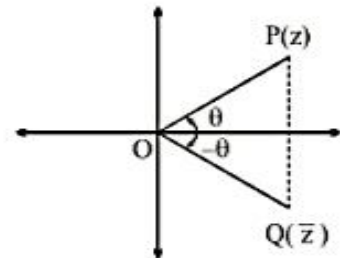
3. THREE IMPORTANT TERMS WITH RESPECT TO COMPLEX NUMBER :

3.1 Conjugate of Complex Number :

Conjugate of a complex number $z = a + ib$ is denoted and defined by $\bar{z} = a - ib$.

In a complex number if we replace i by $-i$, we get conjugate of the complex number. $\bar{z}$ is the mirror image of z about real axis on Argand's Plane.

Geometrical representation of conjugate of complex number



Note :

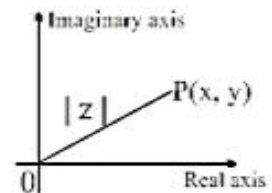
- (a) $z + \bar{z} = 2\text{Re } z$
- (b) $z - \bar{z} = 2i \text{Im } z$
- (c) $z \bar{z} = a^2 + b^2$, where $z = a + ib$
- (d) If z lies in 1st quadrant then $\bar{z}$ lies in 4th quadrant and $-\bar{z}$ in the 2nd Quad.
- (e) If $x + iy = f(a + ib)$ then $x - iy = f(a - ib)$
Further, $g(x + iy) = f(a + ib) \Rightarrow g(x - iy) = f(a - ib)$
e.g. $\sin(\alpha + i\beta) = x + iy \Rightarrow \sin(\alpha - i\beta) = x - iy$

3.2 Modulus of Complex Number :

Modulus of complex number is a distance of the point on the argand plane representing the complex number z from the origin.

If P denotes a complex number $z = x + iy$

$$\text{then } OP = |z| = \sqrt{x^2 + y^2}$$



Note :

- (i) $|z| > 0$.
- (ii) All complex numbers having the same modulus lie on a circle with centre as origin and radius $r = |z|$.

Illustration :

Find the modulus of the following complex numbers

$$(i) \frac{1}{2} + i\frac{\sqrt{3}}{2} \quad (ii) \frac{\sqrt{3}+1}{2\sqrt{2}} - i\frac{\sqrt{3}-1}{2\sqrt{2}} \quad (iii) 1 + \cos \alpha + i \sin \alpha, \alpha \in \left(\pi, \frac{3\pi}{2}\right)$$

Sol. (i) $z = \frac{1}{2} + i\frac{\sqrt{3}}{2}$

$$\therefore |z| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = 1$$

$$(ii) \quad |z| \Rightarrow \sqrt{\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)^2 + \left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)^2} \Rightarrow \sqrt{\frac{3+1+2\sqrt{3}+3+1-2\sqrt{3}}{8}} = \sqrt{1} = 1. \text{ Ans.}$$

$$(iii) \quad z = 1 + \cos \alpha + i \sin \alpha = 2 \cos^2 \frac{\alpha}{2} + 2i \sin \frac{\alpha}{2} \cdot \cos \frac{\alpha}{2} = 2 \cos \frac{\alpha}{2} \left[\cos \frac{\alpha}{2} + i \sin \frac{\alpha}{2} \right]$$

$$|z| = -2 \cos \frac{\alpha}{2} \quad \text{Ans.} \quad \left\{ \frac{\alpha}{2} \in \left(\frac{\pi}{2}, \frac{3\pi}{4} \right) \Rightarrow \cos \frac{\alpha}{2} < 0 \right\}$$

Illustration :

If the complex number z satisfying $z + |z| = 1 + 7i$ then find the value of $|z|^2$.

Sol. $z = x + iy$

$$\therefore x + iy + \sqrt{x^2 + y^2} = 1 + 7i$$

$$x + \sqrt{x^2 + y^2} = 1 \quad \dots (1)$$

and $y = 7 \quad \dots (2)$

$$\therefore x + \sqrt{x^2 + 49} = 1$$

$$x^2 + 49 = 1 + x^2 - 2x$$

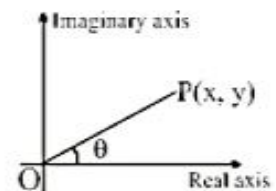
$$2x = -48$$

$$x = -24$$

$$\therefore |z|^2 = x^2 + y^2 = 625 \quad \text{Ans.}$$

3.3 Argument of Complex Number :

Angle (θ) made by the line segment joining the point on the complex plane representing the complex number z to the origin from the positive real axis is called argument of complex number z which is denoted as $\arg(z) = \theta$.



(i) General Argument :

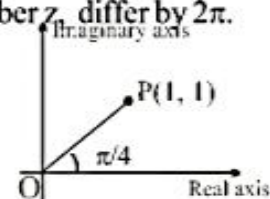
If OP makes an angle θ with real axis then θ is called one of the argument of z .

General values of argument of z are given by

$2n\pi + \theta, n \in \mathbb{I}$. Note that any two argument of the same complex number z , differ by 2π .

e.g. If $z = 1 + i$ then $\arg(z) = \frac{\pi}{4}$

$\therefore$ General value of argument of $z = 2n\pi + \frac{\pi}{4}, n \in \mathbb{I}$



Note that by specifying the modulus and argument, a complex number is completely defined. However for the complex number $0 + 0i$ the argument is not defined and this is the only complex number which is completely defined by talking in terms of its modulus. i.e., $|z| = 0$.

(ii) Amplitude (Principal value of argument) :

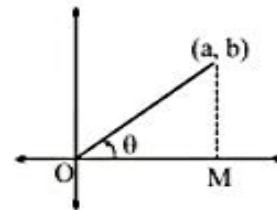
The unique value of θ such that $-\pi < \theta \leq \pi$ is called principal value of argument. Unless otherwise stated, $\text{amp } z$ refers to the principal value of argument.

Working rule for finding principal argument of Complex number Z

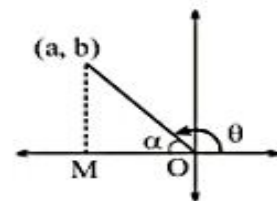
Let $z = a + ib$

First compute $\alpha = \tan^{-1}\left(\frac{|b|}{|a|}\right)$

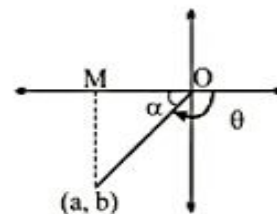
Case I: If z lies in I quadrant i.e. $a, b > 0$
then $\text{amp}(z) = \theta = \alpha$.



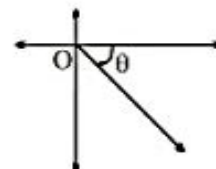
Case II : If z lies in II quadrant i.e. $a < 0, b > 0$
then $\text{amp}(z) = \theta = (\pi - \alpha)$



Case III: If z lies in III quadrant i.e. $a < 0, b < 0$
then $\text{amp}(z) = \theta = -(\pi - \alpha)$



Case IV: If z lies in IV quadrant i.e. $a > 0, b < 0$
then $\text{amp}(z) = \theta = -\alpha$.



Note :

- (i) If z is purely real positive complex number then $\text{amp}(z) = 0$.
- (ii) If z is purely imaginary positive complex number then $\text{amp}(z) = \frac{\pi}{2}$.
- (iii) If z is purely real negative complex number then $\text{amp}(z) = \pi$.
- (iv) If z is purely imaginary neagative complex number then $\text{amp}(z) = \frac{-\pi}{2}$.

Illustration :

Find the amplitude of

(a) $-1 - i\sqrt{3}$ (b) $\frac{1 + \sqrt{3}i}{\sqrt{3} + i}$ (c) $\sin \alpha + i(1 - \cos \alpha), 0 < \alpha < \pi$

Sol.

(a) Let, $z = -1 - i\sqrt{3}$. The $\alpha = \tan^{-1} \left| \frac{b}{a} \right| = \tan^{-1} \left| \frac{\sqrt{3}}{1} \right| = \frac{\pi}{3}$

Clearly, z is in third quadrant.

Therefore argument is $\theta = -(\pi - \alpha) = -\left(\pi - \frac{\pi}{3}\right) = -\frac{2\pi}{3}$.

(b) $\text{amp} \left(\frac{1 + \sqrt{3}i}{\sqrt{3} + i} \right) = \text{amp} \left(\frac{1 + \sqrt{3}i}{\sqrt{3} + i} \times \frac{\sqrt{3} - i}{\sqrt{3} - i} \right) = \text{amp} \left(\frac{2\sqrt{3} + 2i}{4} \right) = \text{amp} \left(\frac{\sqrt{3} + i}{2} \right)$

Complex number $\frac{\sqrt{3} + i}{2}$ lies in 1st quadrant

$\therefore \text{amp}(z) = \theta = \alpha = \frac{\pi}{6}$

(c) $z = \sin \alpha + i(1 - \cos \alpha), 0 < \alpha < \pi;$
 $z = \sin \alpha + i(1 - \cos \alpha)$

$\Rightarrow \text{amp}(z) = \tan^{-1} \left(\frac{1 - \cos \alpha}{\sin \alpha} \right) = \tan^{-1} \left(\frac{2 \sin^2 \frac{\alpha}{2}}{2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2}} \right)$

$= \tan^{-1} \tan \left(\frac{\alpha}{2} \right) = \frac{\alpha}{2} \quad \left\{ 0 < \frac{\alpha}{2} < \frac{\pi}{2} \right\}$

(iii) Least positive argument :

The value of θ such that $0 < \theta \leq 2\pi$ is called the least positive argument.

Illustration :

Find general argument, principal argument and least positive argument of the following complex numbers

- (1) $z_1 = \sqrt{3} - 2i$ (2) $z_2 = -1 + i$ (3) $z_3 = -2 - 3i$
(4) $z_4 = (\sqrt{7} - 2)i$ (5) $z_5 = 2 - \sqrt{7}$ (6) $z_6 = \pi - e$

Sol.

S.No	Complex No.	General Argument	Principal Argument	Least Positive Argument
1.	$\sqrt{3} - 2i$	$2n\pi - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$	$-\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$	$2\pi - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$
2.	$-1 + i$	$2n\pi + \frac{3\pi}{4}$	$\frac{3\pi}{4}$	$\frac{3\pi}{4}$
3.	$-2 - 3i$	$2n\pi - \pi + \tan^{-1}\left(\frac{3}{2}\right)$	$-\left(\pi - \tan^{-1}\left(\frac{3}{2}\right)\right)$	$\pi + \tan^{-1}\left(\frac{3}{2}\right)$
4.	$(\sqrt{7} - 2)i$	$2n\pi + \frac{\pi}{2}$	$\frac{\pi}{2}$	$\frac{\pi}{2}$
5.	$(2 - \sqrt{7})$	$2n\pi + \pi$	π	π
6.	$(\pi - e)$	$2n\pi$	0	2π

Practice Problem

- Q.1 Find the value of $i^n + i^{n+1} + i^{n+2} + i^{n+3}$, $\forall n \in \mathbb{I}$.
- Q.2 If $\frac{(x-2) + (y-3)i}{1+i} = 1 - 3i$ find (x, y).
- Q.3 If $f(x) = x^4 - 4x^3 + 4x^2 + 8x + 44$, find $f(3 + 2i)$.
- Q.4 If $z = \frac{\sqrt{9+40i} + \sqrt{9-40i}}{\sqrt{9+40i} - \sqrt{9-40i}}$, find $|z|$ and z .
- Q.5 A square $P_1P_2P_3P_4$ is drawn in the complex plane with P_1 at (1, 0) and P_3 at (3, 0). Let P_n denotes the point (x_n, y_n) $n = 1, 2, 3, 4$. Find the numerical value of the product of complex numbers $(x_1 + iy_1)(x_2 + iy_2)(x_3 + iy_3)(x_4 + iy_4)$.
- Q.6 Solve the following equations over $\mathbb{C}$ and express the result in the form $a + ib$, $a, b \in \mathbb{R}$.
(a) $ix^2 - 3x - 2i = 0$ (b) $2(1+i)x^2 - 4(2-i)x - 5 - 3i = 0$

- Q.7 Find the sum of the following series
 $i + 2i^2 + 3i^3 + \dots$ to 100 terms.
- Q.8 If $z = (3 + 7i)(p + iq)$ where $p, q \in I - \{0\}$, is purely imaginary then minimum value of $|z|^2$ is
 (A) 0 (B) 58 (C) $\frac{3364}{3}$ (D) 3364
- Q.9 Match the equation in z , in **Column-I** with the corresponding values of $\arg(z)$ in **Column-II**.

Column-I	Column-II
(equations in z)	(principal value of $\arg(z)$)
(A) $z^2 - z + 1 = 0$	(P) $-2\pi/3$
(B) $z^2 + z + 1 = 0$	(Q) $-\pi/3$
(C) $2z^2 + 1 + i\sqrt{3} = 0$	(R) $\pi/3$
(D) $2z^2 + 1 - i\sqrt{3} = 0$	(S) $2\pi/3$

Answer key

- Q.1 0 Q.2 (6, 1) Q.3 5 Q.4 $\frac{\pi}{2}, \frac{4}{5}$ or $-\frac{\pi}{2}, \frac{5}{4}$
- Q.5 15 Q.6 (a) $-i, -2i$; (b) $\frac{3-5i}{2}$ or $-\frac{1+i}{2}$ Q.7 $50(1-i)$
- Q.8 D Q.9 (A) Q, R; (B) P, S; (C) Q, S; (D) P, R

4. REPRESENTATION OF A COMPLEX NUMBER IN DIFFERENT FORMS:

4.1 Cartesian Form / Algebraic Form :

Every complex number expressed in the form of $z = x + iy$ where $x, y \in \mathbb{R}$ and $i = \sqrt{-1}$ is called cartesian form or algebraic form of complex number

for $z = x + iy$, $\operatorname{Re}(z) = x$ and $\operatorname{Im}(z) = y$

$$|z| = \sqrt{x^2 + y^2}, \quad \bar{z} = x - iy, \quad \arg(z) = \tan^{-1}\left(\frac{y}{x}\right)$$

Illustration :

If $\operatorname{Re}\left(\frac{1}{z}\right) > \frac{1}{2}$ then find the locus of z .

Sol. Let $z = x + iy$

$$\frac{1}{z} = \frac{1}{x+iy} = \frac{x-iy}{x^2+y^2} = \frac{x}{x^2+y^2} - \frac{iy}{x^2+y^2}$$

$$\therefore \operatorname{Re}\left(\frac{1}{z}\right) > \frac{1}{2} \Rightarrow \frac{x}{x^2+y^2} > \frac{1}{2}$$

$$\Rightarrow 2x > x^2 + y^2 \Rightarrow x^2 + y^2 - 2x < 0$$

$$\Rightarrow (x-1)^2 + y^2 < 1.$$

Hence, locus of z represents interior region of the circle whose centre is $(1, 0)$ and radius is 1 unit.

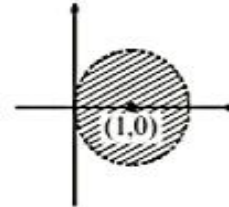


Illustration :

Solve the equation $z^2 + |z| = 0$ where $z \in \mathbb{C}$.

Sol. Let $z = x + iy$
 $z^2 + |z| = 0$

$$x^2 - y^2 + 2xyi + \sqrt{x^2 + y^2} = 0$$

Comparing real and imaginary parts

$$\Rightarrow x^2 - y^2 + \sqrt{x^2 + y^2} = 0 \text{ and } 2xy = 0 \Rightarrow x = 0 \text{ or } y = 0$$

Case-I : When $x = 0$

$$\Rightarrow -y^2 + \sqrt{y^2} = 0 \Rightarrow y^2 = \sqrt{y^2} \Rightarrow y^2 = \pm y$$

$$\Rightarrow y^2 \pm y = 0 \Rightarrow y(y \pm 1) = 0$$

$$y = 0, y = \pm 1$$

$\therefore$ Solutions are $(0, 0)$, $(0, 1)$ and $(0, -1)$

Case-II : When $y = 0$

$$x^2 + \sqrt{x^2} = 0 \Rightarrow x^2 + |x| = 0 \Rightarrow x = 0$$

$\therefore$ Solution is $(0, 0)$ already determined

Hence, solutions of the equation are $(0, 0)$, $(0, 1)$, $(0, -1)$.

Illustration :

If z is a complex number satisfying the equation $|z - (1 + i)|^2 = 2$ and $\omega = \frac{2}{z}$,

then the locus traced by ' ω ' in the complex plane is

(A) $x - y - 1 = 0$ (B) $x + y - 1 = 0$ (C) $x - y + 1 = 0$ (D) $x + y + 1 = 0$

Sol. We have $|z - (1 + i)|^2 = 2$

$$\Rightarrow (x-1)^2 + (y-1)^2 = 2 \quad (\text{Put } z = x + iy)$$

$$\Rightarrow x^2 + y^2 = 2(x + y) \quad \dots\dots(1)$$

$$\text{Let } \omega = h + ik = \frac{2}{z} = \frac{2}{x+iy} = \frac{2(x-iy)}{x^2+y^2}, \text{ so}$$

$$h = \frac{2x}{x^2+y^2}, \quad k = \frac{-2y}{x^2+y^2}$$

$$\Rightarrow \quad h - k = \frac{2(x+y)}{x^2+y^2} = 1 \text{ (from equation (1))}$$

$\therefore$ Locus of the point $\omega(h, k)$ will be $x - y = 1$ Ans. (A)

4.2 Trigonometrical Form / Polar Form :

Let the given complex number be $z = x + iy$
 r and θ be the modulus and $\text{amp}(z)$ respectively.

From the figure $x = r \cos \theta$, $y = r \sin \theta$

$$\therefore \quad z = x + iy = r(\cos \theta + i \sin \theta) = r \text{ cis } \theta$$

Hence, $z = r(\cos \theta + i \sin \theta)$ is called polar / trigonometrical form of the complex number.

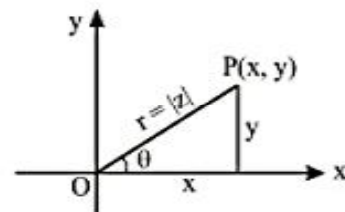


Illustration :

Express the following complex number in polar form.

(i) $z_1 = -1 - i\sqrt{3}$

(ii) $z_2 = -2 + 3i$

Sol.

(i) $z_1 = -1 - i\sqrt{3} = r(\cos \theta + i \sin \theta)$

$$\text{amp}(z_1) = \theta = -\left(\pi - \frac{\pi}{3}\right) = \frac{-2\pi}{3}$$

$$r = \sqrt{1+3} = 2$$

$$\therefore \quad z_1 = -1 - i\sqrt{3} = 2\left(\cos\left(\frac{-2\pi}{3}\right) + i \sin\left(\frac{-2\pi}{3}\right)\right) = 2 \text{ cis } \left(\frac{-2\pi}{3}\right).$$

(ii) $z_2 = -2 + 3i$

$$\text{amp}(z_2) = \theta = \pi - \tan^{-1} \frac{3}{2}$$

$$r = \sqrt{4+9} = \sqrt{13}$$

$$\therefore \quad z_2 = -2 + 3i = \sqrt{13}\left(\cos\left(\pi - \tan^{-1} \frac{3}{2}\right) + i \sin\left(\pi - \tan^{-1} \frac{3}{2}\right)\right) = \sqrt{13} \text{ cis } \left(\pi - \tan^{-1} \frac{3}{2}\right)$$

Illustration :

If $z = 1 + \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5}$ then find modulus and amplitude of z .

Sol. $z = 1 + \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5}$

$$\begin{aligned} & 2 \cos^2 \frac{3\pi}{5} + i 2 \sin \frac{3\pi}{5} \cdot \cos \frac{3\pi}{5} \Rightarrow 2 \cos \frac{3\pi}{5} \left(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} \right) \\ \Rightarrow & 2 \cos \left(\pi - \frac{2\pi}{5} \right) \left(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} \right) \Rightarrow -2 \cos \frac{2\pi}{5} \left(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} \right) \\ \Rightarrow & 2 \cos \frac{2\pi}{5} \left(\cos \left(\pi - \frac{3\pi}{5} \right) - i \sin \left(\pi - \frac{3\pi}{5} \right) \right) \\ \Rightarrow & 2 \cos \frac{2\pi}{5} \left(\cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5} \right) \Rightarrow 2 \cos \frac{2\pi}{5} \left[\cos \left(-\frac{2\pi}{5} \right) + i \sin \left(-\frac{2\pi}{5} \right) \right] \\ \therefore & |z| = 2 \cos \frac{2\pi}{5} \text{ and amp } (z) = -\frac{2\pi}{5}. \end{aligned}$$

4.3 Exponential Form :

$$z = x + iy = r (\cos \theta + i \sin \theta) = r e^{i\theta}$$

$z = r e^{i\theta}$ is called exponential form of the complex number.

where r is modulus of z and θ is amplitude of z .

Here, $\cos \theta + i \sin \theta = e^{i\theta} \forall \theta$ (1)

Replacing i by $-i$, we get

$$\cos \theta - i \sin \theta = e^{-i\theta} \forall \theta \quad \text{.....(2)}$$

Adding (1) and (2)

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \text{ which is purely real}$$

subtracting (2) from (1)

$$i \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2} \text{ which is purely imaginary.}$$

Illustration :

If $z = -2e^{i\left(\frac{-\pi}{3}\right)}$ then find modulus and amplitude of z .

Sol. $z = -2 \left(\cos \left(\frac{-\pi}{3} \right) + i \sin \left(\frac{-\pi}{3} \right) \right)$

$$= -2 \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right) = -2 \left(\frac{1}{2} - i \frac{\sqrt{3}}{2} \right) = -1 + i\sqrt{3} \text{ which lies in } 2^{\text{nd}} \text{ quadrant}$$

modulus $r = \sqrt{1+3} = 2$

$$\text{amp } (z) = \theta = \pi - \tan^{-1} (\sqrt{3}) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}.$$

Illustration :

Find the real part of $z = e^{e^{i\theta}}$, $\theta \in \mathbb{R}$

Sol. $z = e^{e^{i\theta}} = e^{(\cos \theta + i \sin \theta)} = e^{\cos \theta} \cdot e^{i \sin \theta} = e^{\cos \theta} \cdot (\cos(\sin \theta) + i \sin(\sin \theta))$
 $\therefore \operatorname{Re}(z) = e^{\cos \theta} \cdot \cos(\sin \theta)$

4.4 Vectorial Representation of a Complex Number :

Every complex number can be considered as if it is the position vector of that point. If the point P represents the complex number z then,

$$\overrightarrow{OP} = z \quad \& \quad |\overrightarrow{OP}| = |z|.$$

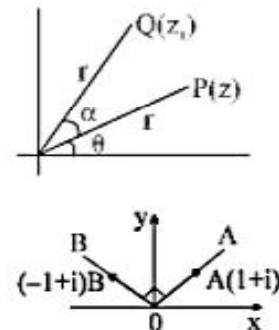
Geometrical meaning of $e^{i\alpha}$.

(i) If $\overrightarrow{OP} = z = r e^{i\theta}$ then $\overrightarrow{OQ} = z_1 = r e^{i(\theta + \alpha)} = z \cdot e^{i\alpha}$.

If $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ are of unequal magnitude then $\overrightarrow{OQ} = \overrightarrow{OP} e^{i\alpha}$

(ii) If $z = \overrightarrow{OA} = 1 + i$ and $\alpha = \frac{\pi}{2}$ then $z_1 = \overrightarrow{OB} = i(1 + i) = -1 + i$

(iii) Using the vectorial concept and section formula complex numbers corresponding to centroid, incentre, orthocentre and circumcentre for a triangle whose vertices are z_1, z_2, z_3 can be deduced.



Practice Problem

- Q.1 Find the locus of z if it satisfies the equation $(z - 1)^2 + |z + 1|^2 = 2$.
 Q.2 Find all the possible solutions of the equation $z^2 = \bar{z}$ where $z \in \mathbb{C}$.
 Q.3 Find the set of all points on the complex plane for which $z^2 + z + 1$ is real and positive.
 Q.4 If $\frac{\pi}{2} < \theta < \frac{3\pi}{2}$ then find modulus and principal argument of $z = (1 + \cos 2\theta) + i \sin 2\theta$.

Answer key

- Q.1 Point $(0, 0)$ Q.2 $(0, 0), (1, 0), \left(\frac{-1}{2}, \frac{\sqrt{3}}{2}\right), \left(\frac{-1}{2}, \frac{-\sqrt{3}}{2}\right)$
 Q.3 $(\lambda, 0), \lambda \in \mathbb{R}$ and $\left(\frac{-1}{2}, y\right)$ where $y \in \left(\frac{-\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$ Q.4 $-2 \cos \theta, \theta - \pi$

5. PROPERTIES OF CONJUGATE, MODULUS AND ARGUMENT :

5.1 Properties of conjugate of complex Numbers :

- (i) $\overline{\overline{z}} = z$ (ii) $|z| = |\overline{z}|$ (iii) $z + \overline{z} = 2\text{Re}(z)$
 (iv) $z - \overline{z} = 2i \text{Im}(z)$ (v) If z is purely real $z = \overline{z}$ (vi) If z is purely imaginary $z = -\overline{z}$
 (vii) $z\overline{z} = |z|^2 = |\overline{z}|^2$
 (viii) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$ In general, $\overline{z_1 + z_2 + \dots + z_n} = \overline{z_1} + \overline{z_2} + \dots + \overline{z_n}$
 (ix) $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$ (x) $\overline{z_1 z_2} = \overline{z_1} \cdot \overline{z_2}$ (xi) $\overline{z^n} = (\overline{z})^n$ (xii) $\overline{\left(\frac{z_1}{z_2}\right)} = \left(\frac{\overline{z_1}}{\overline{z_2}}\right)$
 (xiii) If $\alpha = f(z)$, then $\overline{\alpha} = \overline{f(z)} = f(\overline{z})$ where $\alpha = f(z)$ is a function in complex variable with real coefficients.
 In other words if $f(x + iy) = a + ib$ then $f(x - iy) = a - ib$.

Explanation :

Let $f(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \dots + a_n z^n$, where $a_0, a_1, a_2, \dots, a_n$ are real numbers and z is a complex number. Then

$$f(\overline{z}) = a_0 + a_1 \overline{z} + a_2 (\overline{z})^2 + a_3 (\overline{z})^3 + \dots + a_n (\overline{z})^n = \overline{a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \dots + a_n z^n}.$$

5.2 Properties of modulus of complex numbers :

- (i) $|z| = 0 \Rightarrow z = 0 + i0$
 (ii) $|z| = |-z| = |\overline{z}| = |iz|$
 (iii) $-|z| \leq \text{Re}(z) \leq |z|$ and $-|z| \leq \text{Im}(z) \leq |z|$
 (iv) $z\overline{z} = |z|^2$ If z is unimodular i.e. $|z| = 1$, then $\overline{z} = \frac{1}{z}$
 (v) $|z_1 z_2| = |z_1| |z_2|$ In general $|z_1 \cdot z_2 \cdot z_3 \dots z_n| = |z_1| |z_2| |z_3| \dots |z_n|$
 (vi) $|z^n| = |z|^n$
 (vii) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$ ($z_2 \neq 0$)
 (viii) $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$

Proof:

$$\begin{aligned} |z_1 + z_2|^2 + |z_1 - z_2|^2 &\Rightarrow (z_1 + z_2)(\overline{z_1 + z_2}) + (z_1 - z_2)(\overline{z_1 - z_2}) \\ &\Rightarrow (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) + (z_1 - z_2)(\bar{z}_1 - \bar{z}_2) \\ &\Rightarrow z_1 \bar{z}_1 + z_1 \bar{z}_2 + z_2 \bar{z}_1 + z_2 \bar{z}_2 + z_1 \bar{z}_1 - z_1 \bar{z}_2 - z_2 \bar{z}_1 + z_2 \bar{z}_2 \\ &\Rightarrow 2(|z_1|^2 + |z_2|^2) \end{aligned}$$

OAPB is a parallelogram.

By the vector law of addition

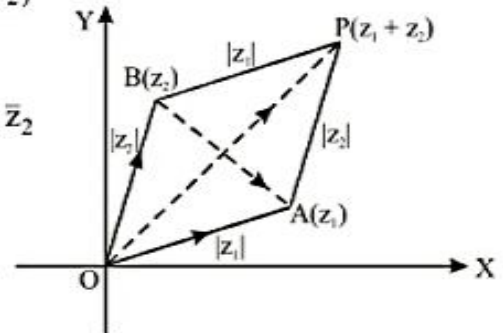
$$\overrightarrow{OP} = z_1 + z_2 \Rightarrow |\overrightarrow{OP}| = |z_1 + z_2|$$

$$\overrightarrow{BA} = z_1 - z_2 \Rightarrow |\overrightarrow{BA}| = |z_1 - z_2|$$

$$\therefore OP^2 + BA^2 = OA^2 + AP^2 + PB^2 + OB^2$$

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$$

The above identity indicates the sum of squares of diagonals of a parallelogram is equal to sum of square of its all four sides.

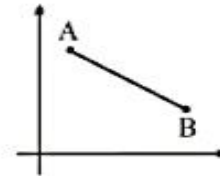


(ix) Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

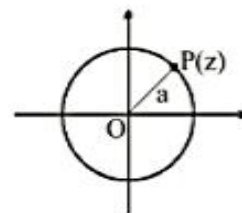
$|z_1 - z_2|$ denotes the distance between two points on the complex plane representing z_1 and z_2 .

$$AB = |z_1 - z_2|$$

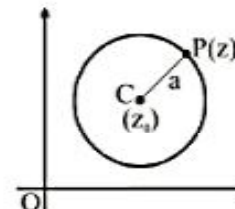
$$\begin{aligned} &= |x_1 + iy_1 - (x_2 + iy_2)| \\ &= |x_1 - x_2 + i(y_1 - y_2)| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \end{aligned}$$



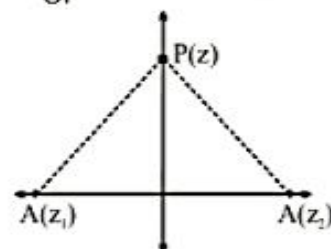
(x) $|z| = a, a \in \mathbb{R}^+ \Rightarrow$ locus of z represents a circle whose centre is origin and radius is 'a'.



(xi) $|z - z_0| = a$ where z_0 is a fixed complex number and $a \in \mathbb{R}^+ \Rightarrow$ locus of z represents a circle whose centre is z_0 and radius is a .



(xii) If $\left| \frac{z - z_1}{z - z_2} \right| = 1$ where z_1 and z_2 are two fixed complex numbers then locus of z is the perpendicular bisector of joining the points representing z_1 and z_2 .



(xiii) Triangle inequalities $||z_1| - |z_2|| \leq |z_1 \pm z_2| \leq |z_1| + |z_2|$

Proof:

Method-I Algebraic Method :

Let $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$

$$z_1 + z_2 = r_1 \cos \theta_1 + r_2 \cos \theta_2 + i (r_1 \sin \theta_1 + r_2 \sin \theta_2)$$

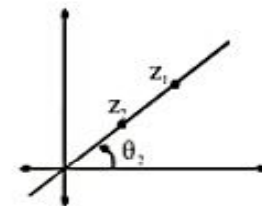
$$|z_1 + z_2| = \sqrt{(r_1 \cos \theta_1 + r_2 \cos \theta_2)^2 + (r_1 \sin \theta_1 + r_2 \sin \theta_2)^2} = \sqrt{r_1^2 + r_2^2 + 2r_1 r_2 \cos(\theta_1 - \theta_2)}.$$

$|z_1 + z_2|$ will be maximum when $\cos(\theta_1 - \theta_2) = 1$

$$\Rightarrow \theta_1 - \theta_2 = 2n\pi$$

$$\Rightarrow \theta_1 = \theta_2 + 2n\pi, n \in \mathbb{I}$$

Hence, for the maximum value of $|z_1 + z_2|$ points representing complex numbers z_1, z_2 and the origin are collinear and z_1, z_2 must lie on the same side of the origin.

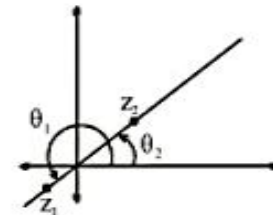


$|z_1 + z_2|$ will be minimum when $\cos(\theta_1 - \theta_2) = -1$

$$\Rightarrow \theta_1 - \theta_2 = 2n\pi + \pi, n \in \mathbb{I}$$

$$\Rightarrow \theta_1 = \theta_2 + 2n\pi + \pi, n \in \mathbb{I}$$

Hence, for the minimum value of $|z_1 + z_2|$ points representing the complex number z_1, z_2 and the origin are collinear and z_1, z_2 must lie on the opposite side of the origin.



In the similar manner minimum and maximum values of $|z_1 - z_2|$ can also be determined.

Method-II Geometrical Method :

Let A and B represent complex numbers z_1 and z_2 respectively.

A parallelogram OAPB is completed $\overrightarrow{OP} = z_1 + z_2$

$$\therefore |\overrightarrow{OP}| = |z_1 + z_2|$$

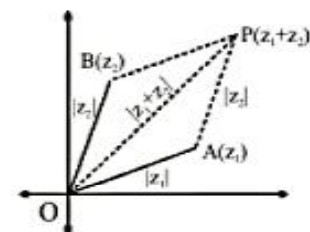
In the $\triangle OAP$, from the fact

(i) Sum of two sides is always greater than third side.

(ii) Absolute value of the difference of two sides is always less than third side.

$$\therefore ||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$$

Note : Sign of equality holds when z_1, z_2 and the origin are collinear.



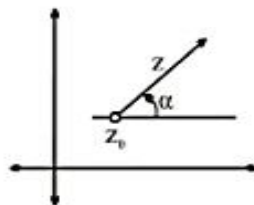
5.3 Properties of Argument of complex number :

- (i) $\text{amp}(z_1 z_2) = \text{amp}(z_1) + \text{amp}(z_2) + 2k\pi, k \in I$
In general $\text{amp}(z_1 \cdot z_2 \dots z_n) = \text{amp}(z_1) + \text{amp}(z_2) + \dots \text{amp}(z_n) + 2k\pi, k \in I$
- (ii) $\text{amp}\left(\frac{z_1}{z_2}\right) = \text{amp}(z_1) - \text{amp}(z_2) + 2k\pi, k \in I$
- (iii) $\text{amp}(z^n) = n \text{amp}(z) + 2k\pi, k \in I$

Note :

In the above properties $2k\pi, k \in I$ is added in RHS and k is chosen in such a way so that value of the expression in RHS belongs to $(-\pi, \pi]$

- (iv) If $\text{amp}(z - z_0) = \alpha$ where z_0 is a fixed complex number then locus of z denotes a ray emanating from z_0 (z_0 is not included) and making an angle α from positive real axis.



Note : For any complex number z

$$\text{amp}(z) + \text{amp}(-\bar{z}) = \pi \quad \text{or} \quad \text{amp}(z) + \text{amp}(-1) + \text{amp}(\bar{z}) = \pi$$

Illustration :

Represent the locus of z satisfying given equation or inequation on the argand's plane.

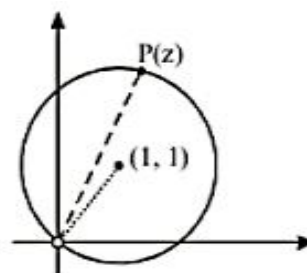
(i) $|z - 1 - i| = \sqrt{2}$ and $\text{amp}(z) = \frac{\pi}{3}$

(ii) $|\text{amp}(z - 2 - i)| < \frac{\pi}{3}$

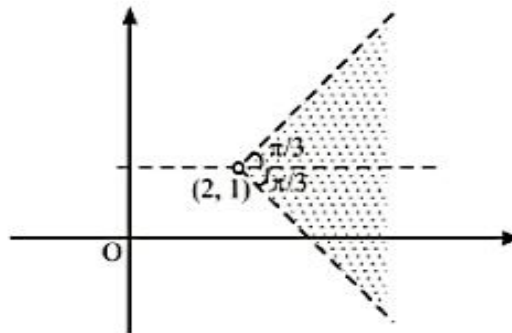
(iii) $3 \leq |z - 4| \leq 4$ and $|\text{amp}(z - 4)| \leq \frac{\pi}{4}$

Sol.

- (i) The complex number z representing the point P is only the solution satisfying both the equations simultaneously



- (ii) Complex numbers corresponding to the points lie in the shaded region satisfy the given inequality.



- (iii) All the complex number z satisfying both the inequalities simultaneously lies in the shaded region

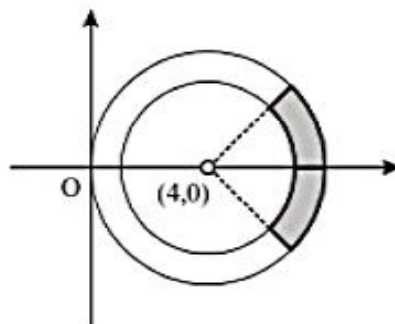


Illustration :

If $|z_1 - 1| \leq 1$, $|z_2 - 2| \leq 2$, $|z_3 - 3| \leq 3$, then find the greatest value of $|z_1 + z_2 + z_3|$.

Sol. $|z_1 + z_2 + z_3| = |(z_1 - 1) + (z_2 - 2) + (z_3 - 3) + 6|$
 $\leq |z_1 - 1| + |z_2 - 2| + |z_3 - 3| + 6 \leq 1 + 2 + 3 + 6 = 12.$

Illustration :

If $|z_1| = |z_2| = |z_3| = 1$, prove that $|z_1 + z_2 + z_3| = \left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right|$.

Sol. We know that $|z| = |\bar{z}|$
 $\Rightarrow |z_1 + z_2 + z_3| = |\bar{z}_1 + \bar{z}_2 + \bar{z}_3|$
 $= \left| \frac{\bar{z}_1 \cdot z_1}{z_1} + \frac{\bar{z}_2 \cdot z_2}{z_2} + \frac{\bar{z}_3 \cdot z_3}{z_3} \right| = \left| \frac{|z_1|^2}{z_1} + \frac{|z_2|^2}{z_2} + \frac{|z_3|^2}{z_3} \right|$
 $= \left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right| \quad (\because |z_1|^2 = |z_2|^2 = |z_3|^2 = 1).$

Illustration :

Consider two pairs of non-zero conjugate complex numbers (z_1, z_2) and (z_3, z_4) .

Find the value of $\arg\left(\frac{z_1}{z_3}\right) + \arg\left(\frac{z_2}{z_4}\right)$.

Sol.
$$\arg\left(\frac{z_1}{z_3}\right) + \arg\left(\frac{z_2}{z_4}\right) = \arg\left(\frac{z_1 z_2}{z_3 z_4}\right)$$

$$= \arg\left(\frac{|z_1|^2}{|z_3|^2}\right) \quad (\text{as } z_2 = \bar{z}_1 \text{ and } z_4 = \bar{z}_3)$$

$$= 0 \quad (\text{as argument of a positive real number is zero}).$$

Illustration :

If z and w are complex numbers satisfying $\bar{z} + i\bar{w} = 0$ and $\text{Amp}(zw) = \pi$, then $\text{Amp}(z)$ is equal to

- (A) $\frac{\pi}{4}$ (B) $\frac{-\pi}{2}$ (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{4}$

Sol. Given $\bar{z} + i\bar{w} = 0 \Rightarrow \bar{z} = -i\bar{w} \quad \text{or} \quad z = iw$

$$\text{amp}(z) - \text{amp}(w) = \text{amp } i = \frac{\pi}{2} \quad \dots(1)$$

also $\text{amp}(zw) = \pi$
 $\text{amp}(z) + \text{amp}(w) = \pi \quad \dots(2)$

(1) + (2), gives $2 \text{amp}(z) = \frac{3\pi}{2} \Rightarrow \text{amp}(z) = \frac{3\pi}{4}$; Also $\text{amp}(w) = \frac{\pi}{4}$

Illustration :

If $\text{Arg}(z+a) = \frac{\pi}{6}$ and $\text{Arg}(z-a) = \frac{2\pi}{3}$; $a \in \mathbb{R}^+$, then

- (A) z is independent of a (B) $|a| = |z+a|$
 (C) $z = a \text{ Cis } \frac{\pi}{6}$ (D) $z = a \text{ Cis } \frac{\pi}{3}$

Sol. Refer the figure z lies on the point of intersection of the rays from A and B . $\triangle ACB$ is a right angle and OBC is an equilateral triangle

$\Rightarrow OC = a \Rightarrow z = a \text{ Cis } \frac{\pi}{3} \Rightarrow (D)$

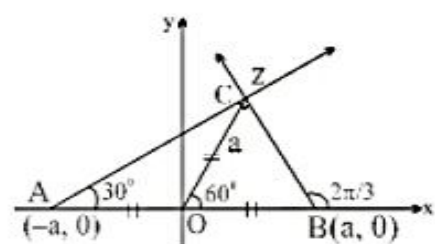


Illustration :

If $|z_1| = 1$, $|z_2| = 2$, $|z_3| = 3$ and $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$,
then find the value of $|z_1 + z_2 + z_3|$.

Sol. $|z_1| = 1 \Rightarrow z_1\bar{z}_1 = 1$, $|z_2| = 2 \Rightarrow z_2\bar{z}_2 = 4$, $|z_3| = 3 \Rightarrow z_3\bar{z}_3 = 9$.

Also, $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$

$$\Rightarrow |z_1z_2z_3\bar{z}_3 + z_1z_2z_3\bar{z}_2 + \bar{z}_1z_2z_3| = 12 \Rightarrow |z_1z_2z_3| |\bar{z}_1 + \bar{z}_2 + \bar{z}_3| = 12$$

$$\Rightarrow |z_1| |z_2| |z_3| |\bar{z}_1 + \bar{z}_2 + \bar{z}_3| = 12 \Rightarrow 6 |\bar{z}_1 + \bar{z}_2 + \bar{z}_3| = 12$$

$$\Rightarrow |\bar{z}_1 + \bar{z}_2 + \bar{z}_3| = 2 \Rightarrow |z_1 + z_2 + z_3| = 2.$$

Illustration :

Find modulus and principal argument of the complex number

$$z = (1+i)(1-i\sqrt{3})(-2-2i)(i)(3).$$

Sol. amp of $(1+i) = \tan^{-1} 1 = \frac{\pi}{4}$

$$\text{amp of } (1-i\sqrt{3}) = -\tan^{-1} |\sqrt{3}| = -\frac{\pi}{3}$$

$$\text{amp of } (-2-2i) = -(\pi - \tan^{-1} 1) = -\frac{3\pi}{4}$$

$$\text{amp of } i = \frac{\pi}{2}$$

$$\text{amp of } 3 = 0$$

$$\text{amp}(z) = \frac{\pi}{4} - \frac{\pi}{3} + \left(-\frac{3\pi}{4}\right) + \frac{\pi}{2} + 0 \Rightarrow -\frac{\pi}{3}$$

$$|z| = |1+i| |1-i\sqrt{3}| |-2-2i| |i| |3|$$

$$= \sqrt{2} \cdot 2 \cdot \sqrt{2} \cdot 1 \cdot 3 = 24. \text{ Ans.}$$

Illustration :

If $|z - (5+7i)| = 9$ then find the greatest and least value of $|z - 2 - 3i|$.

Sol. $|z - 2 - 3i| = |z - (5+7i) + (3+4i)|$

$$||z - (5+7i)| - |3+4i|| \leq |z - (5+7i) + 3+4i| \leq |z - (5+7i)| + |3+4i|$$

$$|9-5| \leq |z - 2 - 3i| \leq 9+5$$

$$4 \leq |z - 2 - 3i| \leq 14. \text{ Ans.}$$

Alternative method :

Z lies on the circle with centre $C(5, 7)$ and radius equal to 9.

Now, we have to find maximum and minimum distance of the point $A(2, 3)$ from the circle.

Maximum distance, $AP = AC + CP = 5 + 9 = 14$

Minimum distance, $AQ = CQ - AQ = 9 - 5 = 4$

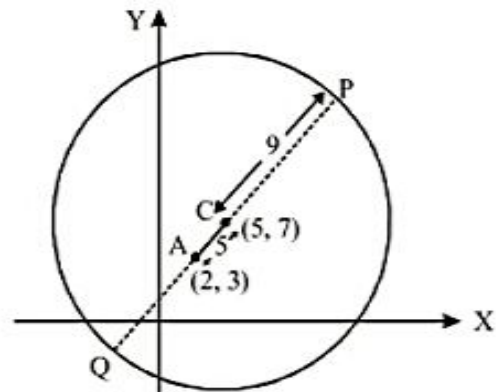


Illustration :

Find the greatest and least value of $|z|$ if z satisfies $\left|z - \frac{4}{z}\right| = 2$.

Sol. $\left|z - \frac{4}{z}\right| \leq \left|z - \frac{4}{z}\right| \leq |z| + \frac{4}{|z|}$

$$\left|z - \frac{4}{z}\right| \leq 2 \leq |z| + \frac{4}{|z|}$$

Put $|z| = r, r > 0$

$$\left|r - \frac{4}{r}\right| \leq 2 \leq r + \frac{4}{r}$$

$$2 \leq r + \frac{4}{r} \Rightarrow r^2 - 2r + 4 \geq 0 \text{ always true} \Rightarrow r > 0$$

$$\left|r - \frac{4}{r}\right| \leq 2$$

$$-2 \leq r - \frac{4}{r} \leq 2$$

$$r^2 + 2r - 4 \geq 0 \Rightarrow (r+1)^2 - 5 \geq 0 \Rightarrow (r+1-\sqrt{5}) \underbrace{(r+1+\sqrt{5})}_{\text{Positive}} \geq 0$$

$$r \geq \sqrt{5} - 1$$

$$r^2 - 2r - 4 \leq 0 \Rightarrow (r-1)^2 - 5 \leq 0 \Rightarrow (r-1-\sqrt{5}) (r-1+\sqrt{5}) \leq 0$$

$$\Rightarrow r \in [1-\sqrt{5}, 1+\sqrt{5}]$$

$$\therefore r \in [\sqrt{5}-1, \sqrt{5}+1] \text{ Ans.}$$

Illustration :

Let z_1 & z_2 be complex numbers such that $z_1 \neq z_2$ and $|z_1| = |z_2|$, then show that $\frac{z_1 + z_2}{z_1 - z_2}$ is purely imaginary.

Sol. $|z_1| = |z_2|$, $\operatorname{Re}\left(\frac{z_1 + z_2}{z_1 - z_2}\right) = \frac{1}{2} \left(\frac{z_1 + z_2}{z_1 - z_2} + \frac{\bar{z}_1 + \bar{z}_2}{\bar{z}_1 - \bar{z}_2} \right) = \frac{1}{2} \left(\frac{2|z_1|^2 - 2|z_2|^2}{|z_1 - z_2|^2} \right) = 0 = \text{purely imaginary.}$

Illustration :

If $\frac{z_1 - 2z_2}{2 - z_1\bar{z}_2}$ is unimodulus and z_2 is not unimodulus, then find $|z_1|$.

Sol. $\left| \frac{z_1 - 2z_2}{2 - z_1\bar{z}_2} \right| = 1 \Rightarrow |z_1 - 2z_2|^2 = |2 - z_1\bar{z}_2|^2$

$$\Rightarrow (z_1 - 2z_2)(\bar{z}_1 - 2\bar{z}_2) = (2 - \bar{z}_1z_2)(2 - z_1\bar{z}_2)$$

$$\Rightarrow z_1\bar{z}_1 - 2\bar{z}_1z_2 - 2z_1\bar{z}_2 + 4z_2\bar{z}_2 = 4 - 2\bar{z}_1z_2 - 2z_1\bar{z}_2 + z_1\bar{z}_1z_2\bar{z}_2$$

$$\Rightarrow |z_1|^2 + 4|z_2|^2 = 4 + |z_1|^2|z_2|^2 \Rightarrow (|z_2|^2 - 1)(|z_1|^2 - 4) = 0$$

$$\Rightarrow |z_1| = 2 \text{ or } |z_2| \neq 1. \text{ Ans.}$$

Illustration :

It is given that complex numbers z_1 and z_2 satisfy $|z_1| = 2$ and $|z_2| = 3$. If the included angle of their corresponding vectors is 60° then $\left| \frac{z_1 + z_2}{z_1 - z_2} \right|$ can be expressed as $\frac{\sqrt{N}}{7}$ where N is natural number then N equals

- (A) 126 (B) 119 (C) 133 (D) 19

Sol. Using cosine rule

$$|z_1 + z_2| = \sqrt{|z_1|^2 + |z_2|^2 - 2|z_1||z_2|\cos 120^\circ}$$

$$= \sqrt{4 + 9 + 2 \cdot 3} = \sqrt{19}$$

and $|z_1 - z_2| = \sqrt{|z_1|^2 + |z_2|^2 - 2|z_1||z_2|\cos 60^\circ}$

$$= \sqrt{4 + 9 - 6} = \sqrt{7}$$

$$\therefore \left| \frac{z_1 + z_2}{z_1 - z_2} \right| = \frac{\sqrt{19}}{\sqrt{7}} = \frac{\sqrt{133}}{7} \Rightarrow N = 133 \text{ Ans.}$$

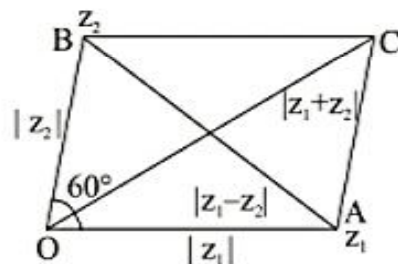


Illustration :

Let $z_1, z_2 \in \mathbb{C}$ be complex numbers such that $|z_1 + z_2| = \sqrt{3}$ and $|z_1| = |z_2| = 1$.
Compute $|z_1 - z_2|$.

Sol. $|z_1 + z_2| = \sqrt{3}$ and $|z_1| = |z_2| = 1$

By squaring

$$(|z_1 + z_2|)^2 = (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) = 3$$

$$= |z_1|^2 + |z_2|^2 + 2\operatorname{Re}(z_1\bar{z}_2) = 3$$

$$\text{or } 2\operatorname{Re}(z_1\bar{z}_2) = 1$$

$$\therefore |z_1 - z_2|^2 = (z_1 - z_2)(\bar{z}_1 - \bar{z}_2)$$

$$= |z_1|^2 + |z_2|^2 - 2\operatorname{Re}(z_1\bar{z}_2) = 1 + 1 - 1$$

$$|z_1 - z_2|^2 = 1$$

$$\therefore |z_1 - z_2| = 1$$

Illustration :

If $z = \frac{1+i\sqrt{3}}{2i\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)}$ then find $|z|$ & $\operatorname{amp}(z)$

Sol. $z = \frac{1+i\sqrt{3}}{(-\sqrt{3}+i)} \Rightarrow |z| = \frac{|1+i\sqrt{3}|}{|-\sqrt{3}+i|} = 1$

$$\operatorname{amp} z = \operatorname{amp}(1+i\sqrt{3}) - \operatorname{amp}(-\sqrt{3}+i) \Rightarrow \frac{\pi}{3} - \frac{5\pi}{6} \Rightarrow \frac{-\pi}{2}. \text{ Ans.}$$

Practice Problem

Q.1 If $\frac{z-1}{z+1}$ is purely imaginary, then prove that $|z| = 1$.

Q.2 Prove that $z = x + iy$ which satisfy the equation $\left|\frac{z-5i}{z+5i}\right| = 1$ lie on the axis of x .

Q.3 If $\arg z = \frac{\pi}{4}$ and $|z+3-i| = 4$ then find z .

Q.4 If z is any complex number such that $|z+4| \leq 3$, then the least value and greatest value of $|z+1|$ are.

Q.5 If $z(2-2\sqrt{3}i)^2 = i(\sqrt{3}+i)^4$, then find the principal argument of z .

Q.6 If $\arg(z) = 50^\circ$ then find principal argument of (z^{100}) .

Q.7 Show that all the roots of the equation $z^n \cos \theta_0 + z^{n-1} \cos \theta_1 + \dots + \cos \theta_n = 2$

where $\theta_0, \theta_1, \theta_2, \dots, \theta_n \in \mathbb{R}$ lies outside the circle $|z| = \frac{1}{2}$

Q.8 Find z if it satisfies $\text{amp}(z+2) = \frac{\pi}{4}$ and $\text{amp}(z-3+2i) = \frac{3\pi}{4}$.

Q.9 If $\text{amp}\left(\frac{(8+i)(7+i)}{5-i}\right) = 6 \tan^{-1}\left(\frac{1}{\lambda}\right)$ then find λ .

Q.10 If $|z_1| = |z_2| = 1$ and $z_1 z_2 \neq -1$ then prove that $\frac{z_1 + z_2}{1 + z_1 z_2}$ is a real number.

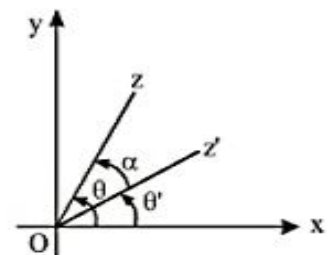
Answer key

Q.3 $z = 1 + i$	Q.4 0, 5	Q.5 $-\frac{\pi}{6}$	Q.6 -40°
Q.8 $-\frac{1}{2} + i\frac{3}{2}$	Q.9 2		

6. CONCEPT OF ROTATION :

If z and z' are two complex numbers then argument of $\frac{z}{z'}$ is the angle through which Oz' must be turned in order that it may lie along Oz .

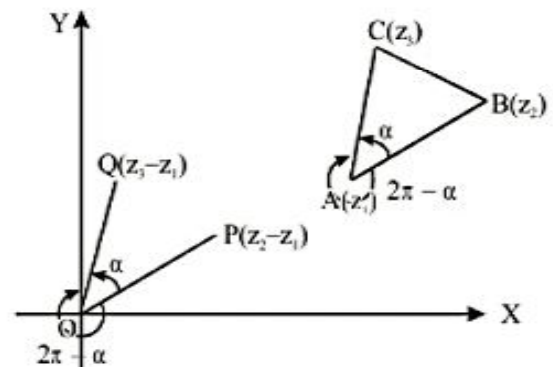
$$\frac{z}{z'} = \frac{|z| e^{i\theta}}{|z'| e^{i\theta'}} = \frac{|z|}{|z'|} e^{i(\theta-\theta')} = \frac{|z|}{|z'|} e^{i\alpha}$$



In general, let z_1, z_2, z_3 be the three vertices of a triangle ABC described in the counter-clockwise sense. Draw OP and OQ parallel and equal to AB and AC respectively. Then the point P is $z_2 - z_1$ and Q is $z_3 - z_1$ and

$$\begin{aligned} \frac{z_3 - z_1}{z_2 - z_1} &= \frac{OQ}{OP} (\cos \alpha + i \sin \alpha) \\ &= \frac{CA}{BA} \cdot e^{i\alpha} = \frac{|z_3 - z_1|}{|z_2 - z_1|} \cdot e^{i\alpha} \end{aligned}$$

Note that $\arg(z_3 - z_1) - \arg(z_2 - z_1) = \alpha$ is the angle through which OP must be rotated in the anti-clockwise direction so that it coincides with OQ.



Here we can write $\frac{z_3 - z_1}{z_2 - z_1} = \frac{|z_3 - z_1|}{|z_2 - z_1|} \cdot e^{-i(2\pi - \alpha)}$ also. In case we are rotating OP in clockwise direction by an angle $(2\pi - \alpha)$. Since the rotation is in clockwise direction, we are taking negative sign with angle $(2\pi - \alpha)$.

Note :

If a complex number (z) is multiplied by i , it means z has been rotated through an angle $\frac{\pi}{2}$ in anticlockwise sense.

e.g $z = 1 + i$

$$z' = (1 + i) e^{i\pi/2} = (1 + i) i = -1 + i.$$

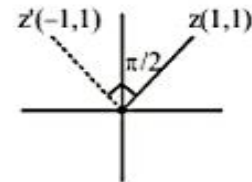
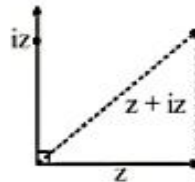
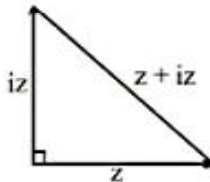


Illustration :

If area of triangle whose sides are represented by z , iz and $z + iz$ is 200 square units then $|z|$ is



Sol. $z = x + iy$ (x, y)
 $iz = ix + i^2y$ ($-y, x$)

$$\frac{1}{2} |z| |iz| = 200$$

$$|z|^2 = 400$$

$$|z| = 20. \text{ Ans.}$$

Illustration :

Consider a square ABCD such that z_1, z_2, z_3 and z_4 represent its vertices A, B, C and D respectively. Express ' z_3 ' and ' z_4 ' in terms of z_1 and z_2 .

Sol. Consider the rotation of AB about A through an angle $\frac{\pi}{4}$. We get

$$\frac{z_3 - z_1}{z_2 - z_1} = \left| \frac{z_3 - z_1}{z_2 - z_1} \right| e^{j\pi/4}$$

$$= \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \Rightarrow z_3 = z_1 + (z_2 - z_1) (1 + i)$$

$$\text{Similarly, } \frac{z_4 - z_1}{z_2 - z_1} = \left| \frac{z_4 - z_1}{z_2 - z_1} \right| e^{j\pi/2} = i$$

$$\Rightarrow z_4 = z_1 + i (z_2 - z_1).$$

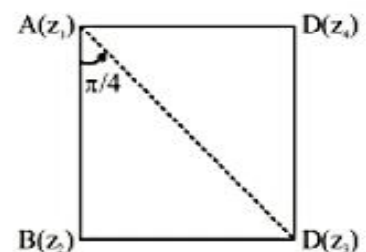


Illustration :

If z_1, z_2, z_3 are the vertices of an equilateral triangle then prove that

$$z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$$

and if z_0 is the circumcentre of the triangle then also prove that $3z_0^2 = z_1^2 + z_2^2 + z_3^2$.

Sol. Since $\triangle ABC$ is equilateral

$$\therefore AB = BC = CA$$

$$|z_1 - z_2| = |z_2 - z_3| = |z_3 - z_1| \quad \dots (1)$$

also $\angle CBA = \angle ACB = \alpha$

Rotate BC about $B(z_2)$ in anticlockwise sense then

$$\Rightarrow \frac{z_1 - z_2}{z_3 - z_2} = \frac{|z_1 - z_2|}{|z_3 - z_2|} \cdot e^{i\alpha} \quad \dots (2)$$

again rotate side AB about $A(z_1)$ in anticlockwise sense then

$$\Rightarrow \frac{z_3 - z_1}{z_2 - z_1} = \frac{|z_3 - z_1|}{|z_2 - z_1|} \cdot e^{i\alpha} \quad \dots (3)$$

By equation (2) and (3)

$$\frac{z_1 - z_2}{z_3 - z_2} = \frac{z_3 - z_1}{z_2 - z_1}$$

$$\Rightarrow z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1 \quad \dots (4)$$

and if z_0 is the circumcentre for equilateral triangle

$$z_0 = \frac{z_1 + z_2 + z_3}{3} \quad \dots (5)$$

or by equation (5)

$$9z_0^2 = z_1^2 + z_2^2 + z_3^2 + 2(z_1 z_2 + z_2 z_3 + z_3 z_1)$$

or by equation (4)

$$z_1^2 + z_2^2 + z_3^2 = 3z_0^2$$

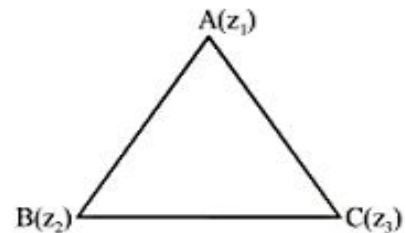


Illustration :

If z_1 & z_2 be non zero complex numbers satisfying the equation, $z_1^2 - 2z_1 z_2 + 2z_2^2 = 0$ then find the geometrical nature of the triangle whose vertices are the origin and the points representing z_1 & z_2

(A) an isosceles right angled triangle

(B) a right angled triangle which is not isosceles

(C) an equilateral triangle

(D) an isosceles triangle which is not right angled.

$$\text{Sol. } \frac{z_1}{z_2} = z \Rightarrow z^2 - 2z + 2 = 0 \Rightarrow z = 1 \pm i$$

$$\Rightarrow \frac{z_1}{z_2} = 1 \pm i \Rightarrow z_1 = z_2 \pm z_2 i \Rightarrow z_1 - z_2 = \pm z_2 i$$

$$\Rightarrow z_1 - z_2 \text{ is perpendicular to } z_2 \text{ and } |z_1 - z_2| = |z_2|$$

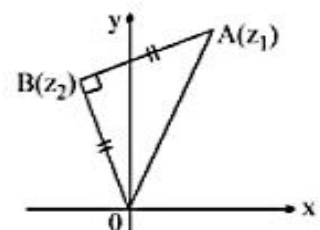


Illustration :

Let z_1 and z_2 be the roots of the equation $z^2 + pz + q = 0$, where the coefficients p and q may be complex number. Let A and B represent z_1 and z_2 in the complex plane. If $\angle AOB = \alpha \neq 0$ and

$OA = OB$, where O is the origin, prove that $p^2 = 4q \cos^2 \left(\frac{\alpha}{2} \right)$.

Sol. $z_1 + z_2 = -p$ (1)

and $z_1 z_2 = q$ (2)

Also $z_2 = z_1 e^{i\alpha}$ (3)

Now $p^2 = (z_1 + z_2)^2 = (z_1 + z_1 e^{i\alpha})^2$
 $= z_1^2 (1 + 2e^{i\alpha} + e^{i2\alpha})$
 $= q e^{-i\alpha} (1 + 2e^{i\alpha} + e^{i2\alpha}) = q(2 + e^{i\alpha} + e^{i\alpha})$
 $= q(2 + 2 \cos \alpha) = 4q \cos^2 \frac{\alpha}{2}.$

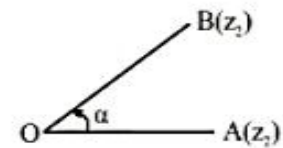
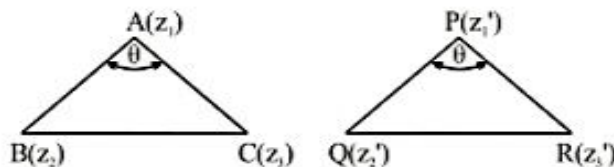


Illustration :

If z_1, z_2, z_3 and z_1', z_2', z_3' represent the vertices of two similar triangles ABC and PQR , respectively then prove that

$$\left| \frac{\bar{z}_1'}{\bar{z}_2 - \bar{z}_1} \right| \left| \frac{z_2 - z_3}{z_3'} \right| + \left| \frac{z_2'}{z_2 - z_1} \right| \left| \frac{\bar{z}_3 - \bar{z}_1}{\bar{z}_3'} \right| \geq 1$$

Sol.



Since ΔABC and ΔPQR are similar, $\frac{AC}{AB} = \frac{PR}{PQ}$ and $\angle BAC = \angle QPR = \theta$.

In ΔABC , $\frac{z_3 - z_1}{z_2 - z_1} = \frac{AC}{AB} e^{i\theta}$ (1)

In ΔPQR , $\frac{z_3' - z_1'}{z_2' - z_1'} = \frac{PR}{PQ} e^{i\theta}$ (2)

$$\begin{aligned}
 &\text{From (1) and (2), } \frac{z_3 - z_1}{z_2 - z_1} = \frac{z_3' - z_1'}{z_2' - z_1'} \\
 &\Rightarrow z_1'(z_2 - z_3) + z_2'(z_3 - z_1) = z_3'(z_2 - z_1) \\
 &\Rightarrow |z_1'(z_2 - z_3) + z_2'(z_3 - z_1)| = |z_3'(z_2 - z_1)| \\
 &\Rightarrow |z_1'(z_2 - z_3)| + |z_2'(z_3 - z_1)| \geq |z_3'(z_2 - z_1)| \\
 &\Rightarrow \left| \frac{z_1'}{z_2 - z_1} \cdot \frac{z_2 - z_3}{z_3'} \right| + \left| \frac{z_2'}{z_2 - z_1} \right| \left| \frac{z_3 - z_1}{z_3'} \right| \geq 1 \\
 &\Rightarrow \left| \frac{z_1'}{z_2 - z_1} \right| \left| \frac{z_2 - z_3}{z_3'} \right| + \left| \frac{z_2'}{z_2 - z_1} \right| \left| \frac{z_3 - z_1}{z_3'} \right| \geq 1 \\
 &\Rightarrow \left| \frac{\bar{z}_1'}{\bar{z}_2 - \bar{z}_1} \right| \left| \frac{z_2 - z_3}{z_3'} \right| + \left| \frac{z_2'}{z_2 - z_1} \right| \left| \frac{\bar{z}_3 - \bar{z}_1}{\bar{z}_3'} \right| \geq 1.
 \end{aligned}$$

Practice Problem

- Q.1 Complex number z_1, z_2, z_3 are the vertices A, B, C respectively of an isosceles right angle triangle with right angle at C . Show that $(z_1 - z_2)^2 = 2(z_1 - z_3)(z_3 - z_2)$
- Q.2 Find the centre of the arc represented by $\arg \left[\frac{(z - 3i)}{(z - 2i + 4)} \right] = \frac{\pi}{4}$.
- Q.3 If z_r ($r = 1, 2, \dots, 6$) are vertices of a regular hexagon and $\sum_{r=1}^6 z_r^2 = k z_0^2$ then find the value of k .
- Q.4 If z_1, z_2, z_3 are collinear then prove that $z_1 |z_2 - z_3| - z_2 |z_3 - z_1| + z_3 |z_1 - z_2| = 0$.
- Q.5 P is a point on the arranged diagram. On the circle with OP an diameter, two points Q & R are taken such that $\angle POQ = \angle QOR = \theta$, if O is the origin and P, Q and R are represented by the complex number z_1, z_2 and z_3 respectively show that $z_2^2 \cos 2\theta = z_1 z_3 \cos^2 \theta$.

Answer key

- Q.2 $\left(\frac{9i - 5}{2} \right)$ Q.3 $k = 6$

7. DEMOIVRE'S THEOREM (DMT) :

Case-I :

Statement :

If n is any integer then

- (i) $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$
 (ii) $(\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2)(\cos \theta_3 + i \sin \theta_3) \dots (\cos \theta_n + i \sin \theta_n)$
 $= \cos (\theta_1 + \theta_2 + \theta_3 + \dots + \theta_n) + i \sin (\theta_1 + \theta_2 + \theta_3 + \dots + \theta_n)$

Case-II :

Statement : If $p, q \in \mathbb{Z}$ and $q \neq 0$ then

$$(\cos \theta + i \sin \theta)^{p/q} = \cos \left(\frac{2k\pi + p\theta}{q} \right) + i \sin \left(\frac{2k\pi + p\theta}{q} \right)$$

where $k = 0, 1, 2, 3, \dots, q-1$

Note : When index ' n ' is integer then $(\cos \theta + i \sin \theta)^n$ has exactly one value which is $\cos n\theta + i \sin n\theta$ but when n is rational number (say $p/q, q \neq 0$) other than integer then $(\cos \theta + i \sin \theta)^{p/q}$ has exactly q different values.

Illustration :

Prove that $\left(\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} \right)^n = \cos n \left(\frac{\pi}{2} - \theta \right) + i \sin n \left(\frac{\pi}{2} - \theta \right).$

Sol. $\frac{(1 + \sin \theta) + i \cos \theta}{(1 + \sin \theta) - i \cos \theta}$

$$\Rightarrow \left[\frac{1 + \cos \left(\frac{\pi}{2} - \theta \right) + i \sin \left(\frac{\pi}{2} - \theta \right)}{1 + \cos \left(\frac{\pi}{2} - \theta \right) - i \sin \left(\frac{\pi}{2} - \theta \right)} \right]^n \Rightarrow \left(\frac{1 + \cos \alpha + i \sin \alpha}{1 + \cos \alpha - i \sin \alpha} \right)^n \text{ where } \alpha = \frac{\pi}{2} - \theta$$

$$\Rightarrow \left(\frac{2 \cos^2 \frac{\alpha}{2} + i 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2}}{2 \cos^2 \frac{\alpha}{2} - i 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2}} \right)^n = \left(\frac{\cos \frac{\alpha}{2} + i \sin \frac{\alpha}{2}}{\cos \frac{\alpha}{2} - i \sin \frac{\alpha}{2}} \right)^n \Rightarrow \left(\frac{e^{i\alpha/2}}{e^{-i\alpha/2}} \right)^n \Rightarrow (e^{i\alpha})^n = e^{i n \alpha}$$

$$\Rightarrow \cos n \left(\frac{\pi}{2} - \theta \right) + i \sin n \left(\frac{\pi}{2} - \theta \right).$$

Aliter : Let $\sin \theta + i \cos \theta = z$

$$\left(\frac{1+z}{1+\bar{z}} \right)^n \Rightarrow \left(\frac{1+z}{1+\frac{1}{z}} \right)^n, \quad \bar{z} = \frac{1}{z} \Rightarrow z^n \Rightarrow (\sin \theta + i \cos \theta)^n$$

$$\Rightarrow \left(\cos \left(\frac{\pi}{2} - \theta \right) + i \sin \left(\frac{\pi}{2} - \theta \right) \right)^n \Rightarrow \cos n \left(\frac{\pi}{2} - \theta \right) + i \sin n \left(\frac{\pi}{2} - \theta \right).$$

Illustration :

If $z = (1+i\sqrt{3})^6 + (1-i\sqrt{3})^6$ the find $|z|$.

Sol. $1 + i\sqrt{3} = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$

$$1 - i\sqrt{3} = 2 \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right)$$

$$\begin{aligned} \therefore z &= (1+i\sqrt{3})^6 + (1-i\sqrt{3})^6 = 2^6 \left[\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)^6 + \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right)^6 \right] \\ &= 2^6 [\cos 2\pi + i \sin 2\pi + \cos 2\pi - i \sin 2\pi] = 2^7 \\ \therefore |z| &= 2^7 \end{aligned}$$

Application of Demoivre's Theorem :

To find roots of a complex quantity is the main application of DMT.

Working rule for finding roots of a complex quantity.

Illustration :

Find all the roots of the equation $z^4 - (1+i) = 0$.

Sol. $z^4 = 1+i \Rightarrow z = (1+i)^{1/4}$

Step-1 : Express $1+i$ in the polar form

$$z = \left(\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right)^{1/4} = 2^{1/8} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)^{1/4}$$

Step-2 : Add $2m\pi$ in the principal argument.

$$z = 2^{1/8} \left(\cos \left(2m\pi + \frac{\pi}{4} \right) + i \sin \left(2m\pi + \frac{\pi}{4} \right) \right)^{1/4}$$

Step-3 : Apply DMT

$$z = z^{1/8} \left(\cos \left(2m\pi + \frac{\pi}{4} \right) \cdot \frac{1}{4} + i \sin \left(2m\pi + \frac{\pi}{4} \right) \cdot \frac{1}{4} \right)$$

Step-4 : Put $m = 0, 1, 2, 3$ to get all four roots of the equation

$$m = 0, z_1 = 2^{1/8} \left(\cos \frac{\pi}{16} + i \sin \frac{\pi}{16} \right), \quad m = 1, z_2 = 2^{1/8} \left(\cos \frac{9\pi}{16} + i \sin \frac{9\pi}{16} \right)$$

$$m = 2, z_3 = 2^{1/8} \left(\cos \frac{17\pi}{16} + i \sin \frac{17\pi}{16} \right), \quad m = 3, z_4 = 2^{1/8} \left(\cos \frac{25\pi}{16} + i \sin \frac{25\pi}{16} \right).$$

7.1 Cube Roots of unity :

$$z^3 - 1 = 0 \Rightarrow z = (1)^{1/3} = (\cos 0 + i \sin 0)^{1/3} = (\cos 2m\pi + i \sin 2m\pi)^{1/3}$$

$$= \cos \frac{2m\pi}{3} + i \sin \frac{2m\pi}{3}, m = 0, 1, 2$$

$$m = 0, z_1 = \cos 0 + i \sin 0 = 1$$

$$m = 1, z_2 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} = \frac{-1}{2} + i \frac{\sqrt{3}}{2} = \omega$$

$$m = 2, z_3 = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} = \frac{-1}{2} - i \frac{\sqrt{3}}{2} = \omega^2$$

Note :

(i) $\omega^3 = 1$

(ii) $\omega^{3n} = 1, n \in \mathbb{I}$

(iii) $1 + \omega + \omega^2 = 0$

(iv) $1 + \omega^r + \omega^{2r} = \begin{cases} 0 & \text{if } r \text{ is not a multiple of } 3. \\ 3 & \text{if } r \text{ is a multiple of } 3. \end{cases}$

(v) Representation of cube roots of unity on argand plane. Cube roots of unity form an equilateral Δ whose side is $\sqrt{3}$ units.

(vi) Some important facts

- $a^3 + b^3 = (a + b)(a^2 - ab + b^2) = (a + b)(a^2 + (\omega + \omega^2)ab + b^2)$
 $= (a + b)[a^2 + ab\omega + ab\omega^2 + b^2\omega^3] = (a + b)[a(a + b\omega) + b\omega^2(a + b\omega)]$
 $\Rightarrow (a + b)(a + b\omega^2)(a + b\omega)$
- $a^3 - b^3 = (a - b)(a - b\omega^2)(a - b\omega)$
- $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$

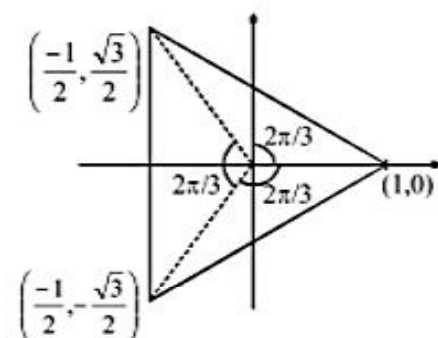


Illustration :

Find the sum of series $1 \cdot (2 - \omega) (2 - \omega^2) + 2 \cdot (3 - \omega) (3 - \omega^2) + \dots + (n - 1)(n - \omega) (n - \omega^2)$ where ω is one of the imaginary cube root of unity.

Sol.

$$\begin{aligned}
 T_n &= (n - 1) (n - \omega) (n - \omega^2) \\
 &= (n - 1) (n^2 - n (\omega + \omega^2) + \omega^3) \\
 &= (n - 1) (n^2 + n + 1) \\
 T_n &= n^3 - 1 \\
 \therefore S_n &= \sum_1^n T_n = \sum n^3 - \sum 1 = \left(\frac{n(n+1)}{2} \right)^2 - n \quad \text{Ans.}
 \end{aligned}$$

Illustration :

If $x = 1 - i\sqrt{3}$, $y = 1 + i\sqrt{3}$ and $z = 2$ then prove that $x^p + y^p = z^p$ where P is a prime number > 3 .

Sol.

$$\begin{aligned}
 x &= (1 - i\sqrt{3}) = -2 \left(\frac{-1 + i\sqrt{3}}{2} \right) = -2\omega \\
 \therefore x^p &= (-2\omega)^p = -(2)^p \omega^p \\
 y &= 1 + i\sqrt{3} = -2 \left(\frac{-1 - i\sqrt{3}}{2} \right) = -2\omega^2 \\
 \therefore y^p &= (-2\omega^2)^p = -(2)^p \omega^{2p} \\
 x^p + y^p &= -2^p (\omega^p + \omega^{2p}) = -(2)^p (-1) = 2^p = z^p
 \end{aligned}$$

Illustration :

If $(a + \omega)^{-1} + (b + \omega)^{-1} + (c + \omega)^{-1} = 2\omega^{-1}$ and $(a + \omega^2)^{-1} + (b + \omega^2)^{-1} + (c + \omega^2)^{-1} = 2\omega^{-2}$ where ω is the complex cube root of unity then show that $(a + 1)^{-1} + (b + 1)^{-1} + (c + 1)^{-1} = 2$. $a, b, c \in \mathbb{R}$.

Sol. Let $\frac{1}{a+x} + \frac{1}{b+x} + \frac{1}{c+x} = \frac{2}{x}$

then $\frac{1}{a+x} + \frac{1}{b+x} = \frac{2}{x} - \frac{1}{c+x}$

$$\Rightarrow x^3 - x(ac + bc + ab) - 2abc = 0 \quad (\text{Let the roots of the equation are } \omega_1, \omega_2 \text{ and } \alpha)$$

If equation has three roots then

sum of roots = 0

$$\therefore \omega + \omega^2 + \alpha = 0 \text{ or } \alpha = 1$$

$$\therefore \frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} = \frac{2}{1}$$

Illustration :

Let z_1 and z_2 be the roots of the equation $x^2 - x + 1 = 0$. Compute

- (i) $x_1^{2000} + x_2^{2000}$ (ii) $x_1^{1999} + x_2^{1999}$ (iii) $x_1^n + x_2^n$ for all $n \in \mathbb{N}$.

Sol. For equation $x^2 - x + 1 = 0$

solutions are $-\omega$ and $-\omega^2$

$$x_1 = -\omega \text{ and } x_2 = -\omega^2$$

$$(i) \quad x_1^{2000} + x_2^{2000} = (-\omega)^{2000} + (-\omega^2)^{2000} \\ \Rightarrow \omega^2 + \omega = -1$$

$$(ii) \quad x_1^{1999} + x_2^{1999} = (-\omega)^{1999} + (-\omega^2)^{1999} \\ \Rightarrow -(\omega + \omega^2) = +1$$

$$(iii) \quad x_1^n + x_2^n = (-\omega)^n + (-\omega^2)^n, n \in \mathbb{N}$$

$$n = 3\lambda \Rightarrow -1 - 1 = -2$$

$$n = 3\lambda + 1 \Rightarrow \omega + \omega^2 = -1$$

$$n = 3\lambda + 2 \Rightarrow \omega^2 + \omega = -1 \text{ or } 1$$

7.2 n , n^{th} root of unity :

$$z^n - 1 = 0 \quad \Rightarrow \quad z = (1)^{1/n} = (\cos 0 + i \sin 0)^{1/n} = (\cos 2m\pi + i \sin 2m\pi)^{1/n}$$

$$\Rightarrow \cos\left(\frac{2m\pi}{n}\right) + i \sin\left(\frac{2m\pi}{n}\right), \quad m = 0, 1, 2, 3, \dots, (n-1)$$

$$m = 0, z_1 = 1$$

$$m = 1, z_2 = \cos\left(\frac{2\pi}{n}\right) + i \sin\left(\frac{2\pi}{n}\right) = e^{i \frac{2\pi}{n}} = \alpha \quad (\text{Let})$$

$$m = 2, z_3 = \cos\left(\frac{4\pi}{n}\right) + i \sin\left(\frac{4\pi}{n}\right) = e^{i \frac{4\pi}{n}} = \alpha^2$$

$\vdots$
 $\vdots$

$$m = n-1, z_n = \cos\left(\frac{2(n-1)\pi}{n}\right) + i \sin\left(\frac{2(n-1)\pi}{n}\right) = e^{i \frac{2(n-1)\pi}{n}} = \alpha^{n-1}.$$

- (1) $1, \alpha, \alpha^2, \alpha^3 + \dots, \alpha^{n-1}$ are n , n^{th} roots of unity which are in G.P. with common ratio α where

$$\alpha = e^{i \frac{2\pi}{n}}.$$

- (2) Sum of n , n^{th} roots of unity is always zero

$$1 + \alpha + \alpha^2 + \alpha^3 + \dots + \alpha^{n-1} = 0$$

Proof: $S = \frac{1 - \alpha^n}{1 - \alpha}$

$$\alpha = e^{\frac{i2\pi}{n}} \Rightarrow \alpha^n = \left(e^{\frac{i2\pi}{n}} \right)^n = e^{i2\pi} = 1$$

$$\therefore S = \frac{1-1}{1-\alpha} = 0$$

(3) Sum of p^{th} powers of n , n^{th} roots of unity = $\begin{cases} n, & \text{if } p \text{ is an integral multiple of } n \\ 0, & \text{if } p \text{ is not an integral multiple of } n \end{cases}$

Proof: $1^p + (\alpha)^p + (\alpha^2)^p + (\alpha^3)^p + \dots + (\alpha^{n-1})^p$

Case-I: When 'p' is an integral multiple of n, $p = n\lambda, \lambda \in \mathbb{I}$

$$\alpha^p = \left(e^{\frac{i2\pi}{n}} \right)^p = e^{\frac{i2\pi}{n}p} = e^{i2\lambda\pi} = 1$$

$$\begin{aligned} \alpha^{2p} &= 1, \alpha^{3p} = 1, \dots, \alpha^{(n-1)p} = 1 \\ \therefore 1 + \alpha^p + \alpha^{2p} + \alpha^{3p} + \dots + \alpha^{(n-1)p} \\ \Rightarrow 1 + 1 + 1 + 1 + \dots + 1 &= n \end{aligned}$$

Case-2 : When p is not an integral multiple of n :

$$\Rightarrow 1 + \alpha^p + \alpha^{2p} + \alpha^{3p} + \dots + \alpha^{(n-1)p}$$

$$= \frac{1 - (\alpha^p)^n}{1 - \alpha^p} \Rightarrow \frac{1-1}{1-\alpha^p} = 0 \quad \{ \because \alpha^{np} = 1 \}$$

Illustration :

If $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ and α_5 are 5, 5th roots of unity then find the value of

(i) $\alpha_1^4 + \alpha_2^4 + \alpha_3^4 + \alpha_4^4 + \alpha_5^4 = 0$
(ii) $\alpha_1^{20} + \alpha_2^{20} + \alpha_3^{20} + \alpha_4^{20} + \alpha_5^{20} = 5$

Sol.

(i) $\because 4$ is not a multiple of 5.

$$\therefore \sum_{i=1}^5 \alpha_i^4 = 0$$

(ii) 20 is a multiple of 5.

$$\therefore \sum_{i=1}^5 \alpha_i^{20} = 5$$

Illustration :

Evaluate :
$$\sum_{\lambda=1}^{12} \left(\sin \frac{2\pi\lambda}{13} - i \cos \frac{2\pi\lambda}{13} \right)$$

Sol.
$$\sum_{\lambda=1}^{12} \left(\sin \frac{2\pi\lambda}{13} - i \cos \frac{2\pi\lambda}{13} \right) \Rightarrow (-i) \sum_{\lambda=1}^{12} \left(\cos \frac{2\pi\lambda}{13} + i \sin \frac{2\pi\lambda}{13} \right)$$

$$\Rightarrow (-i) \sum_{\lambda=1}^{12} e^{\frac{i 2\pi\lambda}{13}} \Rightarrow (-i) \left[1 + e^{\frac{i 2\pi}{13}} + e^{\frac{i 4\pi}{13}} + e^{\frac{i 6\pi}{13}} + \dots + e^{\frac{i 24\pi}{13}} - 1 \right]$$

$$\Rightarrow (-i) (-1) = i. \text{ Ans.}$$

Illustration :

If $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{n-1}$ are non-real n^{th} roots of unity then prove that

(i) $(1 - \alpha_1)(1 - \alpha_2)(1 - \alpha_3) \dots (1 - \alpha_{n-1}) = n$

(ii) $(1 + \alpha_1)(1 + \alpha_2)(1 + \alpha_3) + \dots + (1 + \alpha_{n-1}) = \begin{cases} 0, & \text{if } n \text{ is even} \\ 1, & \text{if } n \text{ is odd} \end{cases}$

Sol.

(i) $z^n - 1 = (z - 1)(z - \alpha_1)(z - \alpha_2)(z - \alpha_3) \dots (z - \alpha_{n-1})$

$$\frac{z^n - 1}{z - 1} = (z - \alpha_1)(z - \alpha_2)(z - \alpha_3) \dots (z - \alpha_{n-1})$$

$$1 + z + z^2 + \dots + z^{n-1} = (z - \alpha_1)(z - \alpha_2) \dots (z - \alpha_{n-1})$$

Put $z = 1$

$$\Rightarrow \underbrace{1 + 1 + 1 + \dots + 1}_{n \text{ times}} = (1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{n-1})$$

Alternative method :

$$\Rightarrow (1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{n-1}) = n$$

$$\lim_{z \rightarrow 1} \frac{z^n - 1}{z - 1} = (1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{n-1})$$

$$\Rightarrow \lim_{z \rightarrow 1} \frac{nz^{n-1} - 1}{1} = (1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{n-1})$$

$$\Rightarrow (1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{n-1}) = n$$

$$(ii) \quad \frac{z^n - 1}{z - 1} = (z - \alpha_1)(z - \alpha_2)(z - \alpha_3) \dots (z - \alpha_{n-1})$$

Put $z = -1$

$$(-1 - \alpha_1)(-1 - \alpha_2)(-1 - \alpha_3) \dots (-1 - \alpha_{n-1}) = \frac{(-1)^n - 1}{-1 - 1}$$

$$\Rightarrow (-1)^{n-1} [(1 + \alpha_1)(1 + \alpha_2) + \dots (1 + \alpha_{n-1})] = \frac{(-1)^n - 1}{-2}$$

$$\Rightarrow (1 + \alpha_1)(1 + \alpha_2) \dots (1 + \alpha_{n-1}) = \frac{(-1)^n - 1}{-2} \cdot \frac{1}{(-1)^{n-1}} \Rightarrow \begin{cases} 0, \text{ even} \\ 1, \text{ odd} \end{cases}$$

Illustration :

If z_1, z_2, z_3, z_4 & z_5 are roots of the equation $z^5 + z^4 + z^3 + z^2 + z + 1 = 0$ then find them also evaluate

$$(i) \sum_{i=1}^5 z_i \quad (ii) \sum_{i=1}^5 z_i^4 \quad (iii) \sum_{i=1}^5 z_i^{12} \quad (iv) \prod_{i=1}^5 (2 - z_i)$$

Sol. $(z^5 + z^4 + z^3 + z^2 + z + 1)(z - 1) = 0$

$$\Rightarrow z^6 - 1 = 0 \Rightarrow z = (1)^{1/6} \Rightarrow z = e^{\frac{i 2m\pi}{6}}, \quad m = 0, 1, 2, \dots, 5$$

Let roots of the equation are $1, z_1, z_2, z_3, \dots, z_5$

$$z_1 = e^{\frac{i2\pi}{6}} = e^{\frac{i\pi}{3}}, \quad z_2 = e^{\frac{i4\pi}{6}} = e^{\frac{i2\pi}{3}}, \quad z_3 = e^{\frac{i6\pi}{6}} = e^{i\pi}, \quad z_4 = e^{\frac{i8\pi}{6}} = e^{\frac{i4\pi}{3}}, \quad z_5 = e^{\frac{i10\pi}{6}} = e^{\frac{i5\pi}{3}}$$

$$(i) \quad 1 + z_1 + z_2 + z_3 + z_4 + z_5 = 0 \Rightarrow \sum_{i=1}^5 z_i = -1$$

$$(ii) \quad 1^4 + z_1^4 + \dots + z_5^4 = 0 \Rightarrow \sum_{i=1}^5 z_i^4 = -1 \quad \{ \because 4 \text{ is not a multiple of } 6 \}$$

$$(iii) \quad \sum_{i=1}^5 z_i^{12}$$

$$\Rightarrow 1 + z_1^{12} + z_2^{12} + \dots + z_5^{12} = 6 \quad \{ \because 12 \text{ is multiple of } 6 \}$$

$$\Rightarrow z_1^{12} + z_2^{12} + \dots + z_5^{12} = 5$$

$$(iv) \prod_{i=1}^5 (2 - z_i)$$

$$z^6 - 1 = (z - 1)(z - z_1)(z - z_2)(z - z_3)(z - z_4)(z - z_5)$$

Put $z = 2$

$$\frac{2^6 - 1}{2 - 1} = (2 - z_1)(2 - z_2) \dots (2 - z_5) \Rightarrow \prod_{i=1}^5 (2 - z_i) = 63 \text{ Ans.}$$

Representation of all the roots of the equation on argand plane :

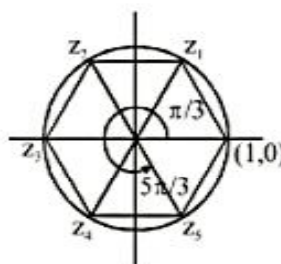


Illustration :

Factorize $z^7 + 1$ into linear and quadratic factors with real coefficients.

Sol. $z^7 + 1 = 0$

$$\Rightarrow z = (-1)^{1/7} = (\cos \pi + i \sin \pi)^{1/7} = (\cos(2m\pi + \pi) + i \sin(2m\pi + \pi))^{1/7}$$

$$= \cos\left(\frac{2m\pi + \pi}{7}\right) + i \sin\left(\frac{2m\pi + \pi}{7}\right) = e^{i\left(\frac{2m\pi + \pi}{7}\right)}, m = 0, 1, 2, \dots, 6$$

$$z_1 = e^{i\pi/7}, z_2 = e^{i3\pi/7}, z_3 = e^{i5\pi/7}, z_4 = e^{i7\pi/7}, z_5 = e^{i9\pi/7}, z_6 = e^{i11\pi/7}, z_7 = e^{i13\pi/7}$$

$$z_7 = e^{i13\pi/7} = e^{-i\pi/7} = \bar{z}_1$$

$$z_6 = e^{i11\pi/7} = e^{-3i\pi/7} = \bar{z}_2$$

$$z_5 = e^{i9\pi/7} = e^{-5i\pi/7} = \bar{z}_3$$

$$\text{Now } z^7 + 1 = (z - z_1)(z - z_2) \dots (z - z_7)$$

$$= (z - z_1)(z - z_2)(z - z_3)(z - z_4)(z - \bar{z}_3)(z - \bar{z}_2)(z - \bar{z}_1)$$

$$= (z^2 - z(z_1 + \bar{z}_1) + |z_1|^2)(z^2 - z(z_2 + \bar{z}_2) + |z_2|^2)(z^2 - z(z_3 + \bar{z}_3) + |z_3|^2)(z - z_4)$$

$$= \left(z^2 - z \cdot 2 \cos \frac{\pi}{7} + 1\right) \left(z^2 - z \cdot 2 \cos \frac{3\pi}{7} + 1\right) \left(z^2 - z \cdot 2 \cos \frac{5\pi}{7} + 1\right) (z + 1)$$

$$= (z + 1) \left(z^2 - 2z \cos \frac{\pi}{7} + 1\right) \left(z^2 - 2z \cos \frac{3\pi}{7} + 1\right) \left(z^2 - 2z \cos \frac{5\pi}{7} + 1\right)$$

Practice Problem

- Q.1 Find the value of the following expression $\left[\frac{1 - \cos \frac{\pi}{10} + i \sin \frac{\pi}{10}}{1 - \cos \frac{\pi}{10} - i \sin \frac{\pi}{10}} \right]^5$.
- Q.2 If $x^2 - x + 1 = 0$ then the value of $\sum_{n=1}^5 \left(x^n + \frac{1}{x^n} \right)^2$.
- Q.3 If α is a non real fifth root of unity, then find the value of $3^{1+\alpha+\alpha^2+\alpha^{-2}-\alpha^{-1}}$ is.
- Q.4 $z_1, z_2, z_3, \dots, z_{n-1}$ are the non real n^{th} roots of unity, then the value of $\frac{1}{(3-z_1)} + \frac{1}{(3-z_2)} + \dots + \frac{1}{(3-z_{n-1})}$
- Q.5 Find the number of roots of the equation $z^{10} - z^5 - 992 = 0$ whose real part is negative.
- Q.6 Least positive argument of the 4th root of the complex number $2 - i\sqrt{12}$ is
(A) $\pi/6$ (B) $5\pi/12$ (C) $7\pi/12$ (D) $11\pi/12$

Answer key

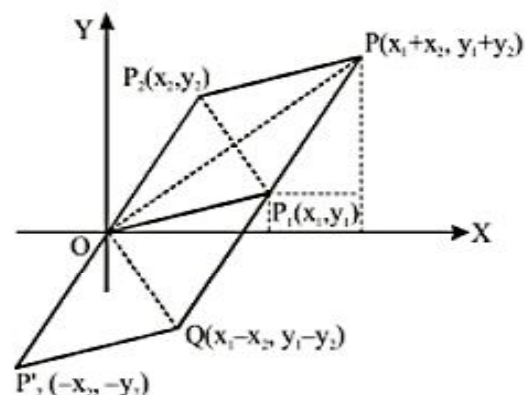
- | | | | | | | | |
|-----|----|-----|---|-----|---|-----|---|
| Q.1 | -1 | Q.2 | 8 | Q.3 | 9 | Q.4 | $\frac{n \cdot 3^{n-1}}{3^n - 1} - \frac{1}{2}$ |
| Q.5 | 5 | Q.6 | B | | | | |

8. GEOMETRY OF COMPLEX NUMBER :

8.1 Geometrical meaning of addition and subtraction :

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be two complex numbers represented by the point $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ respectively. By definition $z_1 + z_2$ should be represented by the point $(x_1 + x_2, y_1 + y_2)$. This point is the vertex which completes the parallelogram with the line segments joining the origin with OP_1 and OP_2 as the adjacent sides.

$$\Rightarrow |z_1 + z_2| = OP$$



Also by definition $z_1 - z_2$ should be represented by the point $(x_1 - x_2, y_1 - y_2)$. This point is the vertex which completes the parallelogram with the line segments joining the origin with OP_1 and OP'_2 (where the point P'_2 represents $-z_2$; the point $-z_2$ can be obtained by producing the directed line P_2O by length $(|z_2|)$ as the adjacent sides.

$$\Rightarrow |z_1 - z_2| = OQ = P_2P_1$$

8.2 Geometrical Meaning of product and Division :

Let $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$, $z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$ be complex numbers represented by Q_1 and Q_2 .

(i) Construction for the point representing the product $z_1 z_2$:

Let L be the point on OX which represents unity, so that $OL = 1$. Draw the triangle OQ_2P directly similar to the triangle OLQ_1 . Then point P represents the product $z_1 z_2$.

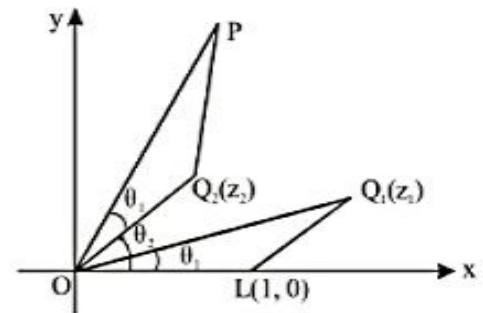
Explanation :

Due to similar triangles

$$\frac{OP}{OQ_1} = \frac{OQ_2}{OL}, \text{ that is } \frac{OP}{r_1} = \frac{r_2}{1} \Rightarrow OP = r_1 r_2$$

Also $\angle Q_2OP = \angle LOQ_1 = \theta_1 \Rightarrow \angle LOP = \theta_1 + \theta_2$

Since $z_1 z_2 = r_1 r_2 \{ \cos (\theta_1 + \theta_2) + i \sin (\theta_1 + \theta_2) \}$, P represents $z_1 z_2$.



(ii) Construction for the point representing the quotient z_1/z_2 :

Draw the triangle OQ_1P directly similar to the triangle OQ_2L

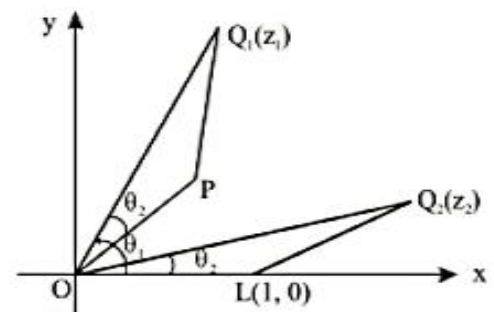
Then P represents the quotient $\frac{z_1}{z_2}$

Explanation :

From the last construction,

$$\frac{OQ_1}{OQ_2} = \frac{OP}{OL} \Rightarrow \frac{r_1}{r_2} = \frac{OP}{1}$$

number represented by $P \cdot z_2 = z_1 \Rightarrow$ number represented by $P = \frac{z_1}{z_2}$



Remark :

(i) If $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$, and $z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$,

then $z_1 z_2 = r_1 r_2 = e^{i(\theta_1 + \theta_2)}$ and $\frac{z_1}{z_2} = \frac{r_1}{r_2} \cdot e^{i(\theta_1 - \theta_2)}$

(ii) $\arg(z_1 z_2) = \theta_1 + \theta_2 = \arg(z_1) + \arg(z_2)$

(iii) $\arg\left(\frac{z_1}{z_2}\right) = \theta_1 - \theta_2 = \arg(z_1) - \arg(z_2)$

8.3 Centroid, Incentre, Orthocentre & Circumcentre of a triangle on a complex plane :

- (i) Centroid 'G' = $\frac{z_1 + z_2 + z_3}{3}$
 (ii) Incentre 'I' = $\frac{a z_1 + b z_2 + c z_3}{a + b + c}$
 (iii) Orthocentre :

$$Z_D = \frac{b \cos C z_2 + c \cos B z_3}{a}$$

Now $AE = c \cos A$;

$$l = AE \operatorname{cosec} C = c \cos A \operatorname{cosec} C$$

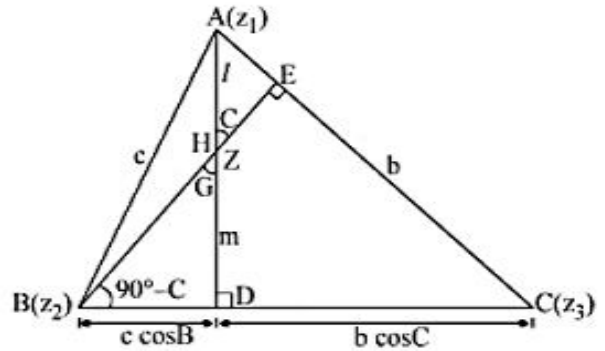
$$l = 2R \cos A \quad \left(\frac{c}{\sin C} = 2R \right)$$

and $m = c \cos B \cot C$ or $m = 2R \cos B \cos C$

Hence $Z_{II} = \frac{m z_1 + l Z_D}{l + m}$

$$\begin{aligned} &= \frac{2R \cos B \cos C z_1 + 2R \cos A \left(\frac{b \cos C z_2 + c \cos B z_1}{a} \right)}{2R (\cos A + \cos B \cos C)} \\ &= \frac{a \cos B \cos C z_1 + b \cos A \cos C z_2 + c \cos A \cos B z_3}{a (-\cos(B+C) + \cos B \cos C)} \\ &= \frac{z_1 (\sin A \cos B \cos C) + (\sin B \cos C \cos A) z_2 + (\sin C \cos A \cos B) z_3}{\sin A (\sin B \sin C)} \end{aligned}$$

$$Z_H = \frac{z_1 \tan A + z_2 \tan B + z_3 \tan C}{\Sigma \tan A} \quad \text{or} \quad \frac{z_1 \tan A + z_2 \tan B + z_3 \tan C}{\Pi \tan A}$$



(iv). Circumcentre :

We have z_0 being equidistant from the vertices gives,

$$|z_1 - z_0| = |z_2 - z_0| = |z_3 - z_0|$$

Consider, $|z_1 - z_0|^2 = |z_2 - z_0|^2$

$$(z_1 - z_0)(\bar{z}_1 - \bar{z}_0) = (z_2 - z_0)(\bar{z}_2 - \bar{z}_0)$$

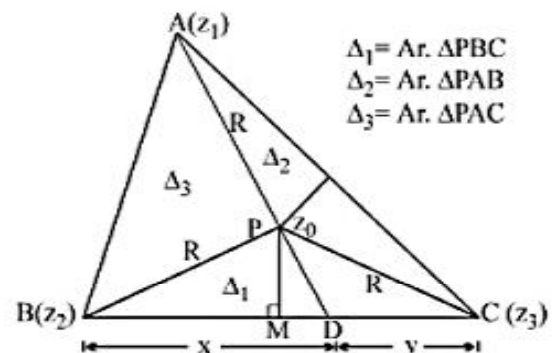
$$\bar{z}_1(z_1 - z_0) - \bar{z}_2(z_2 - z_0) = \bar{z}_0[(z_1 - z_0) - (z_2 - z_0)]$$

$$\bar{z}_1(z_1 - z_0) - \bar{z}_2(z_2 - z_0) = \bar{z}_0(z_1 - z_2) \quad \dots(1)$$

Similarly 1st & 3rd gives

$$\bar{z}_1(z_1 - z_0) - \bar{z}_3(z_3 - z_0) = \bar{z}_0(z_1 - z_3) \quad \dots(2)$$

dividing (1) by (2) eliminate $\bar{z}_0$ and get z_0 .



8.4 Different forms of equation of a straight line :

(i) Equation of straight line with the help of coordinate geometry :

Writing $x = \frac{z + \bar{z}}{2}$, $y = \frac{z - \bar{z}}{2i}$ etc. in $\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$ and re-arranging terms, we find that the equation of the line through z_1 and z_2 is given by.

$$\frac{z - z_1}{z_2 - z_1} = \frac{\bar{z} - \bar{z}_1}{\bar{z}_2 - \bar{z}_1} \quad \text{or} \quad \begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$$

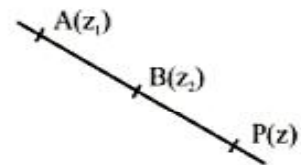
(ii) Equation of a straight line with the help of rotation formula :

Let $A(z_1)$ and $B(z_2)$ be any two points lying on any line and we have to obtain the equation of this line. For this purpose let us take any point

$P(z)$ lying on this line. Since $\arg\left(\frac{z - z_1}{z_2 - z_1}\right) = 0 \text{ or } \pi$,

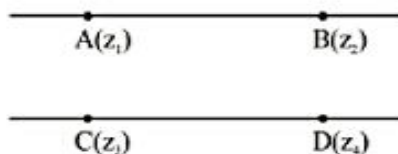
$\frac{z - z_1}{z_2 - z_1}$ is purely real.

$$\Rightarrow \frac{z - z_1}{z_2 - z_1} = \frac{\bar{z} - \bar{z}_1}{\bar{z}_2 - \bar{z}_1}$$



Condition for which two lines which are parallel or perpendicular :

(a) For parallel :



Since AB and CD lines are parallel

$$\therefore \arg\left(\frac{z_1 - z_2}{z_3 - z_4}\right) = 0 \text{ or } \pi \Rightarrow \frac{z_1 - z_2}{z_3 - z_4} \text{ is purely real} \Rightarrow \boxed{\frac{z_1 - z_2}{z_3 - z_4} = \frac{\bar{z}_1 - \bar{z}_2}{\bar{z}_3 - \bar{z}_4}}$$

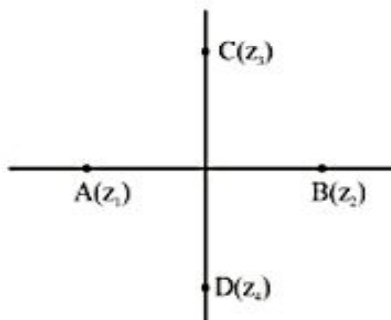
(b) For perpendicular :

Since lines AB and CD are perpendicular

$$\arg\left(\frac{z_1 - z_2}{z_3 - z_4}\right) = \pm \frac{\pi}{2}$$

$$\Rightarrow \frac{z_1 - z_2}{z_3 - z_4} \text{ is purely imaginary}$$

$$\Rightarrow \boxed{\frac{z_1 - z_2}{z_3 - z_4} = -\left(\frac{\bar{z}_1 - \bar{z}_2}{\bar{z}_3 - \bar{z}_4}\right)}$$



(c) **General equation of the line :**

From equation (i) we get, $z(\bar{z}_2 - \bar{z}_1) - z_1 \bar{z}_2 + z_1 \bar{z}_1 = \bar{z}(z_2 - z_1) - \bar{z}_1 z_2 + z_1 \bar{z}_1$

$$\Rightarrow z(\bar{z}_2 - \bar{z}_1) + \bar{z}(\bar{z}_1 - \bar{z}_2) + \bar{z}_1 z_2 - z_1 \bar{z}_2 = 0$$

Here $\bar{z}_1 z_2 - z_1 \bar{z}_2$ is a purely imaginary number as $\bar{z}_1 z_2 - z_1 \bar{z}_2 = 2i \operatorname{Im}(\bar{z}_1 z_2)$.

Let $\bar{z}_1 z_2 - z_1 \bar{z}_2 = ib, \quad b \in \mathbb{R}$

$$\Rightarrow z(\bar{z}_2 - \bar{z}_1) + \bar{z}(\bar{z}_1 - \bar{z}_2) + ib = 0 \Rightarrow z i(\bar{z}_1 - \bar{z}_2) + \bar{z} i(z_2 - z_1) + b = 0$$

Let $a = i(z_2 - z_1) \Rightarrow \bar{a} = i(\bar{z}_1 - \bar{z}_2)$

$$\Rightarrow \bar{a}z + a\bar{z} + b = 0$$

This is the general equation of a line in the complex plane.

8.5 Complex slope of a line :

(a) Complex slope of a line $\bar{a}z + a\bar{z} + b = 0$ is defined as

$$\omega = -\frac{a}{\bar{a}}$$

(b) Complex slope of a line passing through $A(z_1)$ & $B(z_2)$ is given by

$$\omega = \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2}$$

(c) Complex slope of a line making an angle θ from positive real axis is given by

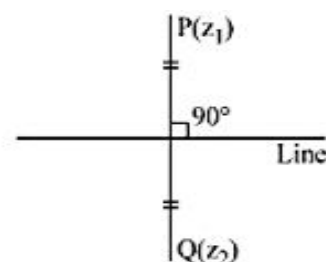
$$\omega = e^{i2\theta}$$

Note : Let ω_1 and ω_2 be the complex slopes of two lines

- (i) If $\omega_1 = \omega_2$ then lines are parallel
- (ii) If $\omega_1 + \omega_2 = 0$ then lines are perpendicular.

8.6 Reflection Points For A Line (Image of a point in a line) :

Two given points P & Q denoted by complex numbers z_1 and z_2 respectively are the reflection points in a given straight $\bar{\alpha}z + \alpha\bar{z} + r = 0$ if the given line is the right bisector of the line segment PQ . Two points $P(z_1)$ & $Q(z_2)$ will be the reflection points of each other in the given straight line, iff $\bar{\alpha}z_1 + \alpha\bar{z}_2 + r = 0$ where 'r' is real and α is non zero complex constant.



Proof: M is the mid point of PQ which lies on the line mirror

$$\therefore \bar{\alpha} \left(\frac{z_1 + z_2}{2} \right) + \alpha \left(\frac{\bar{z}_1 + \bar{z}_2}{2} \right) + r = 0$$

$$\bar{\alpha} z_1 + \bar{\alpha} z_2 + \alpha \bar{z}_1 + \alpha \bar{z}_2 + 2r = 0 \quad \dots(i)$$

Line PQ and the given line both are perpendicular

$$\therefore \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2} + \left(-\frac{\alpha}{\bar{\alpha}} \right) = 0$$

$$z_1 \bar{\alpha} - z_2 \bar{\alpha} = \alpha \bar{z}_1 - \alpha \bar{z}_2$$

$$\alpha \bar{z}_1 + \bar{\alpha} z_2 = \alpha \bar{z}_2 + \bar{\alpha} z_1 \quad \dots(ii)$$

From (i) & (ii)

$$2(\bar{\alpha} z_2 + \alpha \bar{z}_1) + 2r = 0$$

$$\Rightarrow \alpha \bar{z}_1 + \bar{\alpha} z_2 + r = 0$$

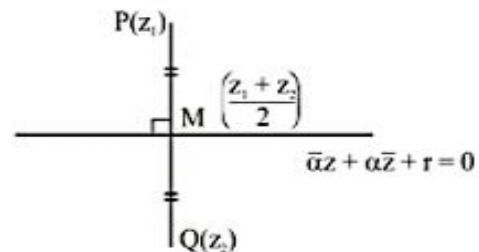


Illustration :

Find the image of the point $P(1 - i)$ in the line mirror $(1 + i)z - (1 - i)\bar{z} + 2i = 0$

Sol. The given line mirror is

$$(1 + i)z - (1 - i)\bar{z} + 2i = 0$$

$$(-1 + i)z - (1 + i)\bar{z} - 2 = 0$$

$$(1 - i)z + (1 + i)\bar{z} + 2 = 0$$

Let z_2 be the image of the given point $P(1 - i)$

$$\Rightarrow (1 - i)z_2 + (1 + i)(1 + i) + 2 = 0 \quad \Rightarrow (1 - i)z_2 + 1 - 1 + 2i + 2 = 0$$

$$\Rightarrow z_2 = \frac{-2(1 + i)}{1 - i} = \frac{-2(1 - 1 + 2i)}{2} = -2i$$

8.7 Equation of perpendicular bisector :

Consider a line segment joining $A(z_1)$ and $B(z_2)$. Let the line 'L' be it's perpendicular bisector.

If $P(z)$ be any point on the 'L'. we have

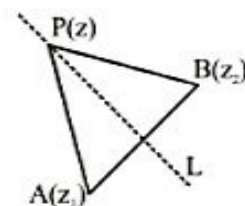
$$PA = PB \Rightarrow |z - z_1| = |z - z_2|$$

$$\Rightarrow |z - z_1|^2 = |z - z_2|^2$$

$$\Rightarrow (z - z_1)(\bar{z} - \bar{z}_1) = (z - z_2)(\bar{z} - \bar{z}_2)$$

$$\Rightarrow z\bar{z} - z\bar{z}_1 - \bar{z}_1\bar{z} + z\bar{z} - z\bar{z}_2 - \bar{z}_2\bar{z} + z\bar{z}_2$$

$$\Rightarrow z(\bar{z}_2 - \bar{z}_1) + \bar{z}(\bar{z}_2 - \bar{z}_1) + z_1\bar{z}_1 - z_2\bar{z}_2 = 0$$



8.8 Distance of a given point from a given line :

Let the given line be $z\bar{a} + \bar{z}a + b = 0$, and the given point be z_c

Say $z_c = x_c + iy_c$.

Replacing z by $x + iy$, in the given equation,

we get, $x(a + \bar{a}) + iy(\bar{a} - a) + b = 0$

Distance of (x_c, y_c) from this line is,

$$\frac{|x_c(a + \bar{a}) + iy_c(\bar{a} - a) + b|}{\sqrt{(a + \bar{a})^2 - (a - \bar{a})^2}} = \frac{|z_c\bar{a} + \bar{z}_c a + b|}{\sqrt{4(\operatorname{Re}(a))^2 + 4(\operatorname{Im}(a))^2}} = \frac{|z_c\bar{a} + \bar{z}_c a + b|}{2|a|}$$

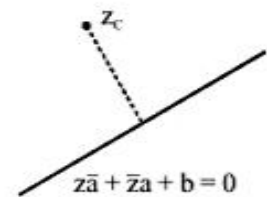


Illustration :

Consider a line $(1 + i)z - (1 - i)\bar{z} + 2i = 0$. Find the distance of a point $2i$ from the above line.

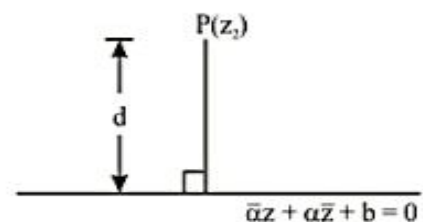
Sol. The given line can be written as

$$(1 - i)z + (1 + i)\bar{z} + 2 = 0$$

Applying distance formula

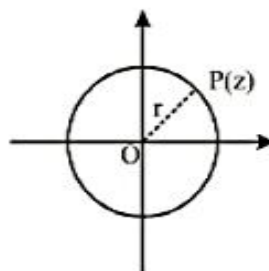
$$d = \frac{|z_c\bar{\alpha} + \bar{z}_c\alpha + b|}{2|\alpha|}$$

$$= \left| \frac{(1 - i)2i + (1 + i)(-2i) + 2}{2\sqrt{2}} \right| = \left| \frac{2i + 2 - 2i + 2 + 2}{2\sqrt{2}} \right| = \frac{3}{\sqrt{2}} \text{ Ans.}$$

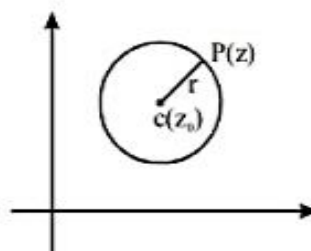


8.9 Different forms of equation of a circle :

(i) $|z| = r$, $r \in \mathbb{R}^+$ then locus of z represent a circle whose centre is the origin and radius is equal to r .



(ii) $|z - z_0| = r$, $r \in \mathbb{R}^+$ then locus of z represents a circle whose centre is z_0 and radius is equal to r .



- (iii) Equation $z\bar{z} + a\bar{z} + \bar{a}z + b = 0$ represents a circle whose centre is $-a$ and radius is equal $\sqrt{|a|^2 - b}$.

Equation of the circle with centre z_0 and radius 'r'.

$$|z - z_0| = r \Rightarrow |z - z_0|^2 = r^2 \Rightarrow (z - z_0)(\bar{z} - \bar{z}_0) = r^2 \Rightarrow z\bar{z} - z\bar{z}_0 - \bar{z}z_0 + |z_0|^2 - r^2 = 0$$

Putting $-z_0 = a$ and $|z_0|^2 - r^2 = b$ equation becomes

$$z\bar{z} + \bar{a}z + a\bar{z} + b = 0$$

$$\therefore \text{Centre } z_0 = -a \text{ and radius } r = \sqrt{|z_0|^2 - b} = \sqrt{|a|^2 - b}$$

Illustration

Find the centre and radius of the circle

$$z\bar{z} - (3 - 4i)z - (3 + 4i)\bar{z} + 9 = 0$$

$$z\bar{z} + (-3 + 4i)z + (-3 - 4i)\bar{z} + 9 = 0$$

Sol. Here, $a = -3 - 4i$, $b = 9$

$\therefore$ centre, $-a = 3 + 4i$ or $(3, 4)$

$$\text{radius } r = \sqrt{|a|^2 - b} = \sqrt{|-3 - 4i|^2 - 9} = \sqrt{25 - 9} = 4$$

- (iv) **Diametric form of equation of circle :**

Let $A(z_1)$ and $B(z_2)$ are the extremities of diameter of a circle and $P(z)$ be a variable point then

- (a) $CP = r$ (r = radius of the circle)

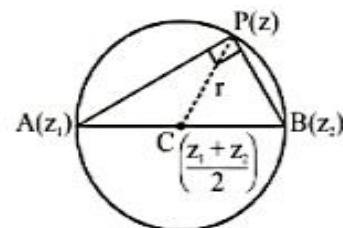
$$\left| z - \frac{z_1 + z_2}{2} \right| = \left| \frac{z_1 - z_2}{2} \right|$$

- (b) AP & BP are perpendicular

$$\therefore \omega_1 + \omega_2 = 0 \Rightarrow \frac{z - z_1}{\bar{z} - \bar{z}_1} + \frac{z - z_2}{\bar{z} - \bar{z}_2} = 0$$

- (c) $AP^2 + BP^2 = AB^2$

$$|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$$



- (v) **Equation of circle passing through three non-collinear points :**

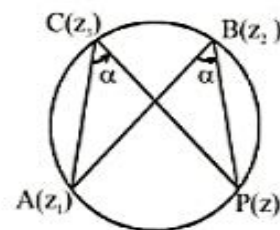
Let $A(z_1)$, $B(z_2)$ and $C(z_3)$ be three given non-collinear points and $P(z)$ be a variable point.

Angle subtended by the arc AP at B and C are equal (say ' α ')

Using rotation theorem :

$$\frac{z - z_2}{z_1 - z_2} = \lambda_1 e^{i\alpha} \quad \dots(i)$$

$$\frac{z - z_3}{z_1 - z_3} = \lambda_2 e^{i\alpha} \quad \dots(ii)$$



(i) $\div$ (ii)

$$\frac{z-z_2}{z_1-z_2} \cdot \frac{z_1-z_3}{z-z_3} = \frac{\lambda_1}{\lambda_2} = \text{Purely real}$$

$$\therefore \left(\frac{z-z_2}{z-z_3} \right) \left(\frac{z_1-z_3}{z_1-z_2} \right) = \left(\frac{\bar{z}-\bar{z}_2}{\bar{z}-\bar{z}_3} \right) \left(\frac{\bar{z}_1-\bar{z}_3}{\bar{z}_1-\bar{z}_2} \right)$$

The above equation represents a circle passing through $A(z_1)$, $B(z_2)$ and $C(z_3)$.

Imp. Note :

Let z_1 and z_2 be two given complex numbers and z be any complex number

such that, $\arg \left(\frac{z-z_1}{z-z_2} \right) = \alpha$, where $\alpha \in (0, \pi)$

Then 'z' would lie on an arc of segment of a circle on

$z_1 z_2$, containing angle α . Clearly if $\alpha \in \left(0, \frac{\pi}{2} \right)$, z would

lie on the major arc (excluding the points z_1 and z_2) and

if $\alpha \in \left(\frac{\pi}{2}, \pi \right)$, 'z' would lie on the minor arc

(excluding the points z_1 and z_2).

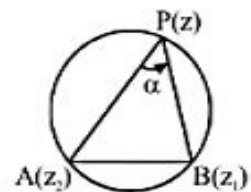


Figure-I

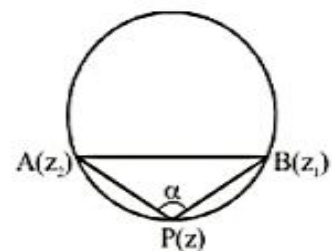


Illustration :

If $z_1 = 2 + 3i$, $z_2 = 3 - 2i$ and $z_3 = -1 - 2\sqrt{3}i$ then which of the following is true?

(A) $\arg \left(\frac{z_3}{z_2} \right) = \arg \left(\frac{z_3 - z_1}{z_2 - z_1} \right)$

(B) $\arg \left(\frac{z_3}{z_2} \right) = \arg \left(\frac{z_2}{z_1} \right)$

(C) $\arg \left(\frac{z_3}{z_2} \right) = 2 \arg \left(\frac{z_3 - z_1}{z_2 - z_1} \right)$

(D) $\arg \left(\frac{z_3}{z_2} \right) = \frac{1}{2} \arg \left(\frac{z_3 - z_1}{z_2 - z_1} \right)$

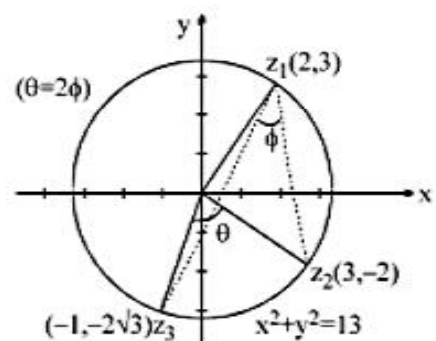
Sol. Note that $|z_1| = |z_2| = |z_3| = \sqrt{13}$

Hence z_1, z_2, z_3 lies on a circle with centre $(0, 0)$

and $r = \sqrt{13}$ as shown

now $\arg \frac{z_2}{z_3} = 2 \arg \frac{z_2 - z_1}{z_3 - z_1}$

$\therefore \arg \frac{z_3}{z_2} = 2 \arg \frac{z_3 - z_1}{z_2 - z_1} \Rightarrow (C)$



8.10 Condition of concyclic of four points :

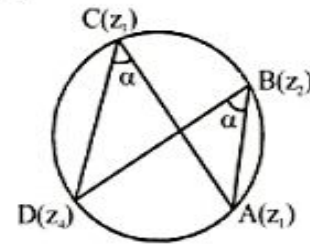
Four points $A(z_1)$, $B(z_2)$, $C(z_3)$ and $D(z_4)$ taken in order are cyclic then

$$\frac{z_1 - z_2}{z_4 - z_2} = \lambda_1 e^{i\alpha} \quad \dots(i)$$

$$\frac{z_1 - z_3}{z_4 - z_3} = \lambda_2 e^{i\alpha} \quad \dots(ii)$$

From (i) & (ii)

$$\frac{z_1 - z_2}{z_4 - z_2} \cdot \frac{z_4 - z_3}{z_1 - z_3} = \text{Purely real (Which is the required condition)}$$



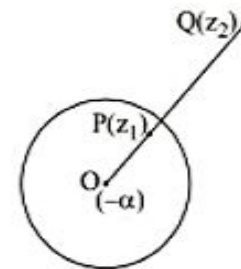
8.11 Inverse Point w.r.t. a circle :

Two points $P(z_1)$ & $Q(z_2)$ are said to be the inverse point of each other w.r.t. the circle if

- (i) O, P, Q are collinear and lie on the same side of the centre 'O' and
- (ii) $(OP) \cdot (OQ) = \rho^2$ when ρ is the radius of the circle.

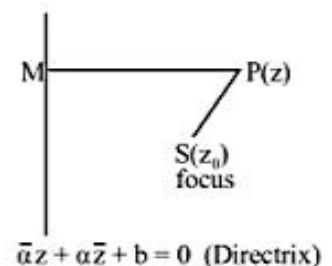
The points z_1 & z_2 will be the inverse point of each other w.r.t. the circle $z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + r = 0$ if

$$z_1\bar{z}_2 + \bar{\alpha}z_1 + \alpha\bar{z}_2 + r = 0$$



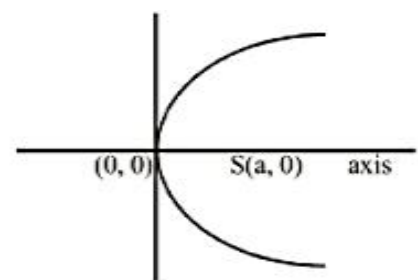
8.12 General loci on complex plane :

- (a) (i) If $|z - z_0| = \left| \frac{\bar{\alpha}z + \alpha\bar{z} + b}{2|\alpha|} \right|$ then locus of z is a parabola whose focus is z_0 and directrix is the line $\bar{\alpha}z + \alpha\bar{z} + b = 0$ provided $\bar{\alpha}z_0 + \alpha\bar{z}_0 + b \neq 0$



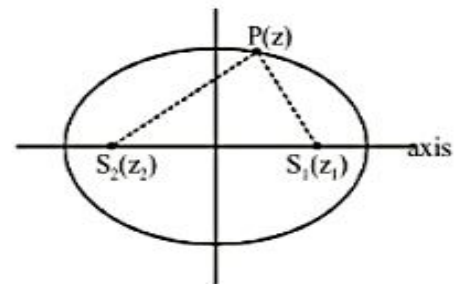
- (ii) If $(z - \bar{z})^2 + 8a(z + \bar{z}) = 0$ then locus of z represents a parabola whose focus is $(a, 0)$, vertex is $(0, 0)$ and real axis is the axis of parabola.

$$\begin{aligned} \text{Put } z &= x + iy \\ \Rightarrow (i2y)^2 + 8a.2x &= 0 \\ \Rightarrow -4y^2 + 16ax &= 0 \\ \Rightarrow y^2 &= 4ax \end{aligned}$$



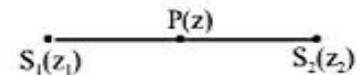
(b) $|z - z_1| + |z - z_2| = k$

(i) $k > |z_1 - z_2|$ then locus of z represents an ellipse whose foci are z_1 and z_2 .



(ii) $k = |z_1 - z_2|$ then locus of z is line segment joining z_1 and z_2 .

$PS_1 + PS_2 = S_1S_2 = |z_1 - z_2| \Rightarrow P$ lies on the line segment joining $S_1(z_1)$ and $S_2(z_2)$.



(c) $||z - z_1| - |z - z_2|| = k$

(i) $k < |z_1 - z_2|$ then locus of z represents a hyperbola whose foci are z_1 and z_2 .

$$|PS_1 - PS_2| < S_1S_2$$

$$||z - z_1| - |z - z_2|| < |z_1 - z_2|$$

(ii) $k = |z_1 - z_2|$ then locus of z is union of two rays emanating from z_1 and z_2 .

$$|PS_1 - PS_2| = S_1S_2$$

$$||z - z_1| - |z - z_2|| = |z_1 - z_2|$$

$\therefore P(z)$ lies on the rays emanating either z_1 or z_2 .

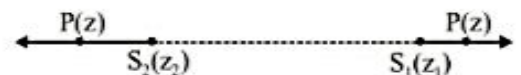
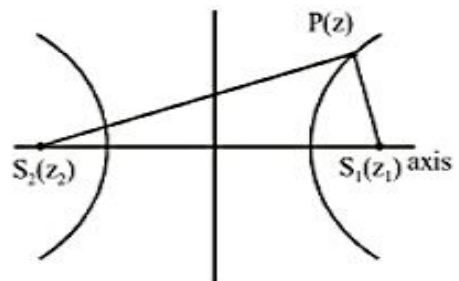


Illustration :

If z is a complex number satisfying the equation $|z + i| + |z - i| = 8$, on the complex plane then maximum value of $|z|$ is

(A) 2

(B) 4

(C) 6

(D) 8

Sol. If $|z + i| + |z - i| = 8$,

$$PF_1 + PF_2 = 8$$

$$\therefore |z|_{\max} = 4 \Rightarrow (B)$$

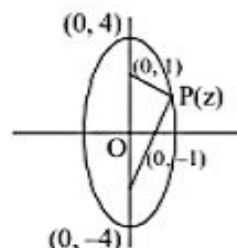


Illustration :

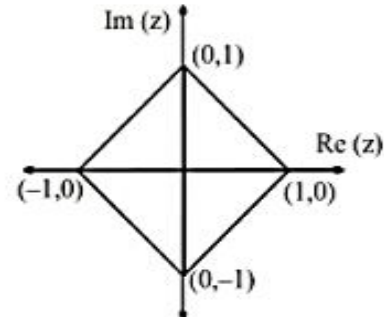
Number of complex numbers satisfying the relation $|z + \bar{z}| + |z - \bar{z}| = 2$ and $|z + i| + |z - i| = 2$, is

- (A) 1 (B) 2 (C) 3 (D) 4

Sol. We have $\left| \frac{z + \bar{z}}{2} \right| + \left| \frac{z - \bar{z}}{2} \right| = 1 \Rightarrow |x| + |y| = 1$

Also, $|z - i| + |z + i| = 2$
 $\Rightarrow$ A line segment between $(0, 1)$ and $(0, -1)$.

So, number of solution is 2
i.e., $z = i$ and $-i$



9. LOGARITHM OF COMPLEX QUANTITY :

Let u and z be two complex number such that
 $u = e^z$ where $u = \alpha + i\beta$ and $z = x + iy$ or $\alpha + i\beta = e^{x+iy}$
then z is called the logarithm of u to the base e

$$z = \log_e u$$

or $x + iy = \log_e (\alpha + i\beta)$

Now $\alpha + i\beta = e^{x+iy} = e^x \cdot e^{iy} = e^x (\cos y + i \sin y)$

Hence corresponding to some values of x and y there is one and only one value of $\alpha + i\beta$.

Now we will prove that for a given value of u , there can be infinite values of z .

Here $z = \log u$

or $x + iy = \log (\alpha + i\beta)$
 $= \log r (\cos \theta + i \sin \theta)$

[where $\alpha = r \cos \theta$, $\beta = r \sin \theta$ and $r = \sqrt{(\alpha^2 + \beta^2)}$, $\theta = \tan^{-1} \beta/\alpha$]

$$= \log r \{ \cos (\theta + 2n\pi) + i \sin (\theta + 2n\pi) \}$$

$$= \log r e^{i(\theta + 2n\pi)}, \text{ where } n \in \mathbb{I}$$

$$= \log r + i(\theta + 2n\pi)$$

$$= \log \sqrt{(\alpha^2 + \beta^2)} + i(2n\pi + \tan^{-1} \beta/\alpha) \dots (1)$$

From (1) it is clear that for different values of n , we have different values of $x + iy$.

Hence logarithm of a complex quantity is a multi-valued function and it is expressed as $\text{Log } u$ i.e.,

$$\text{Log } u = \text{Log } (\alpha + i\beta) = \frac{1}{2} \log(\alpha^2 + \beta^2) + i \tan^{-1} \beta/\alpha + 2n\pi i$$

If we put $n = 0$, we obtain the principal value of $\log u$.

Hence the principal value of $\log u = \frac{1}{2} \log(\alpha^2 + \beta^2) + i\theta$ and $\theta (= \tan^{-1} \beta/\alpha)$ should be such that $-\pi < \theta \leq \pi$.

Illustration :

Separate into real and imaginary parts the following :

(i) $\log(1+i)$ (ii) $\log(-5)$

Sol.

(i) $\log(1+i) = \log(1+i) + 2n\pi i$

Let $1 = r \cos \theta$ and $1 = r \sin \theta$

then $r = \sqrt{1^2 + 1^2} = \sqrt{2}$

and $\tan \theta = 1$ or $\theta = \pi/4$

$$\begin{aligned} \therefore \log(1+i) &= \log(r \cos \theta + ir \sin \theta) \\ &= \log r (\cos \theta + i \sin \theta) \\ &= \log \sqrt{2} (\cos \pi/4 + i \sin \pi/4) \\ &= \log(\sqrt{2} e^{i\pi/4}) \\ &= \log \sqrt{2} + \log e^{i\pi/4} \\ &= \frac{1}{2} \log 2 + \frac{1}{4} i\pi \end{aligned}$$

Here $\text{Log}(1+i) = \frac{1}{2} \log 2 + \frac{1}{4} i\pi + 2n\pi i$

$$= \frac{1}{2} \log 2 + i(2n\pi + \pi/4)$$

(ii) $\text{Log}(-5) = \log(-5) + 2n\pi i$

Now $\log(-5) = \log(-5 + i.0)$

$$= \log(r \cos \theta + ir \sin \theta)$$

where $r \cos \theta = -5, r \sin \theta = 0,$

$\therefore r = \sqrt{(-5)^2 + 0^2} = 5, r \sin \theta = 0,$

$\therefore \log(-5) = \log 5 (\cos \pi + i \sin \pi)$

$$= \log(5e^{i\pi})$$

$$= \log 5 + i\pi$$

Hence $\text{Log}(-5) = \log 5 + i\pi + 2n\pi i$

$$= \log 5 + i(2n+1)\pi$$

Illustration :

If $\tan \log(x+iy) = a+ib$ and $a^2+b^2 \neq 1$, then prove that

$$\tan \log(x^2+y^2) = \frac{2a}{1-a^2-b^2}.$$

Sol. Here $\tan \log(x+iy) = a+ib$

$\therefore \log(x+iy) = \tan^{-1}(a+ib) \quad \dots (1)$

and $\log(x-iy) = \tan^{-1}(a-ib) \quad \dots (2)$

$\therefore \log(x+iy) + \log(x-iy) = \tan^{-1}(a+ib) + \tan^{-1}(a-ib)$

$$\text{or} \quad \log \left\{ (x+iy)(x-iy) = \tan^{-1} \left\{ \frac{(a+ib)+(a-ib)}{1-(a+ib)(a-ib)} \right\} \right\}$$

$$\text{or} \quad \log (x^2 + y^2) = \tan^{-1} \left\{ \frac{2a}{1-a^2-b^2} \right\}$$

$$\text{or} \quad \tan \log (x^2 + y^2) = \frac{2a}{1-a^2-b^2}$$

Illustration :

Prove that : $i^i = \exp \{-(4n+1)\pi/2\}$

$$\begin{aligned} \text{Sol.} \quad i^i &= \exp (i \log i) \\ &= \exp \{i (\log i + 2n\pi i)\} \\ &= \exp [i \{2n\pi i + \log (0 + 1.i)\}] \\ &= \exp [i \{2n\pi i + \log (\cos \pi/2 + i \sin \pi/2)\}] \\ &= \exp [i (2n\pi i + \log e^{i\pi/2})] \\ &= \exp \{i (2n\pi i + i\pi/2)\} = \exp (-2n\pi - \pi/2) \\ &= \exp \{-(4n+1)\pi/2\} \end{aligned}$$

where $n = 0, 1, 2, 3, \dots$

Illustration :

If $i^{i-i\infty} = A + iB$ (principal values only being considered then prove that)

$$\tan \left(\frac{\pi A}{2} \right) = \frac{B}{A} \quad \text{and} \quad A^2 + B^2 = e^{-B\pi}$$

$$\begin{aligned} \text{Sol.} \quad \text{Here } i^{i-i\infty} &= A + iB \\ \therefore i^{A+iB} &= A + iB \\ \Rightarrow (A + iB) \log \{ \cos (\pi/2) + i \sin (\pi/2) \} &= \log (A + iB) \\ \Rightarrow (A + iB) \log e^{i\pi/2} &= \frac{1}{2} \log (A^2 + B^2) + i \tan^{-1} (B/A) \\ \Rightarrow (A + iB) (i\pi/2) &= \frac{1}{2} \log (A^2 + B^2) + i \tan^{-1} (B/A) \end{aligned}$$

Equating real and imaginary parts of both sides

$$\begin{aligned} -\left(\frac{B\pi}{2} \right) &= \frac{1}{2} \log (A^2 + B^2) \\ \Rightarrow A^2 + B^2 &= e^{-B\pi} \\ \text{and } A \pi/2 &= \tan^{-1} (B/A) \Rightarrow B/A = \tan (A\pi/2) \\ \Rightarrow \tan \frac{A\pi}{2} &= \frac{B}{A}. \end{aligned}$$

Hence proved.

Practice Problem

- Q.1 Find the equation of the line passing through $1 + i$ and $2 + i$ on argand plane and also find the complex numbers corresponding to the points on the line which are at a distance of $\sqrt{3}$ units from $1 + i \cdot 0$.
- Q.2 Find the equation of a line passing through $A(z_1)$ and perpendicular to OA where O is the origin.
- Q.3 Locate the complex number $z = x + iy$ which satisfies the inequality $\log_{\sqrt{3}} \frac{|z^2| - |z| + 1}{2 + |z|} < 2$.
- Q.4 If A & B represent the complex numbers z_1 and z_2 such that $|z_1 + z_2| = |z_1 - z_2|$ then find the circumcentre of $\triangle OAB$, where O is the origin.
- Q.5 Locate the points representing complex number z for which
- (i) $\frac{\pi}{3} < \arg z \leq \frac{3\pi}{2}$ (ii) $\arg\left(\frac{z-1-i}{z-2}\right) = \frac{\pi}{3}$
- Q.6 Find the equation of a circle which touches the line $iz + \bar{z} + 1 + i = 0$ and the lines $(z-i)z = (2+i)\bar{z}$ and $(2+i)z + (i-2)\bar{z} - 4i = 0$ are the normals to the circle.
- Q.7 If $|z+2| + |z| \leq 8$, then the range of values of $|z-4|$.
- Q.8 Prove the following
- (a) $\text{Log}(-i) = 2n\pi i - \frac{1}{2}i\pi$ (b) $\log(-i) = -\frac{1}{2}\pi i$ (c) $\log_i i = \frac{4m+1}{4n+1}$, $m, n \in i$
- Q.9 $i \log \frac{x-i}{x+i} = \pi - 2 \tan^{-1} x$

Answer key

- Q.1 $1 + \sqrt{2} + i, 1 - \sqrt{2} + i$ Q.2 $z\bar{z}_1 + z_1\bar{z} - 2|z_1|^2 = 0$
- Q.3 z lies in the interior region of the circle whose centre is the origin and radius is equal to 5.
- Q.4 $\frac{z_1 + z_2}{2}$
- Q.6 $|z - (1 - 2i)| = 2\sqrt{2}$ Q.7 $[1, 9]$

SOLVED EXAMPLES

- Q.1 The value of $169 e^{i\left(\pi + \sin^{-1} \frac{12}{13} + \cos^{-1} \frac{5}{13}\right)}$ is
(A) $119 - 120i$ (B) $120 + 119i$ (C) $119 + 120i$ (D) None of these

Sol. $169 e^{i\left(\pi + \sin^{-1} \frac{12}{13} + \cos^{-1} \frac{5}{13}\right)}$

$$= -169 \left[\cos\left(\cos^{-1} \frac{5}{13}\right) + i \sin\left(\cos^{-1} \frac{5}{13}\right) \right] \left[\cos\left(\sin^{-1} \frac{12}{13}\right) + i \sin\left(\sin^{-1} \frac{12}{13}\right) \right]$$

$$= -169 \left[\frac{5}{13} + i \frac{12}{13} \right] \left[\frac{5}{13} + i \frac{12}{13} \right]$$

$$= [119 - 120i] = -i [120 + 119i]$$

Hence (A) is the correct answer.

- Q.2 If $A(z_1)$, $B(z_2)$ and $C(z_3)$ be the vertices of a triangle ABC in which $\angle ABC = \frac{\pi}{4}$ and $\frac{AB}{BC} = \sqrt{2}$, then the value of z_2 is equal are
(A) $z_3 + i(z_1 + z_3)$ (B) $z_3 - i(z_1 - z_3)$ (C) $z_3 + i(z_1 - z_3)$ (D) None of these

Sol. $\frac{AB}{BC} = \sqrt{2}$

Considering the rotation about B, we get

$$\frac{z_1 - z_2}{z_3 - z_2} = \left| \frac{z_1 - z_2}{z_3 - z_2} \right| e^{i\pi/4} = \frac{AB}{BC} e^{i\pi/4}$$

$$\Rightarrow \sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right) = 1 + i \Rightarrow z_1 - z_2 = \{1 + i\} (z_3 - z_2)$$

$$\Rightarrow z_1 - (1 + i)z_3 = z_2(1 - 1 - i) = -iz_3$$

$$\Rightarrow z_2 = -z_3(z_3 + z_1)$$

- Q.3 If $f(x) = g(x^3) + xh(x^3)$ is divisible by $x^2 + x + 1$, then
(A) $g(x)$ is divisible by $(x - 1)$ but not by $h(x)$ (B) $h(x)$ is divisible by $(x - 1)$ but not by $g(x)$
(C) both $g(x)$ and $h(x)$ are divisible by $(x - 1)$ (D) None of these

Sol. $f(x) = g(x^3) + xh(x^3)$
Let $f_1(x) = 1 + x + x^2$
Clearly, the roots of $f_1(x) = 0$ are ω and ω^2 (where ω is a non-real cube root of unity). As $f_1(x)$ divides $f(x)$.

$\Rightarrow f(\omega) = 0, f(\omega^2) = 0 \Rightarrow g(\omega^3) + \omega h(\omega^3) = 0$ and $g(\omega^6) + \omega^2 h(\omega^6) = 0$
 $\Rightarrow g(1) + \omega h(1) = 0, g(1) + \omega^2 h(1) = 0 \Rightarrow 2g(1) + h(1)(\omega + \omega^2) = 0$
 $\Rightarrow 2g(1) - h(1) = 0 \Rightarrow h(1) = 2g(1)$
 $\Rightarrow g(1) + \omega \cdot 2g(1) = 0 \Rightarrow g(1)(1 + 2\omega) = 0 \Rightarrow g(1) = 0 \Rightarrow g(1) = 0$
 $\Rightarrow x = 1$ is the root of $g(x) = 0$ and $h(x) = 0$. Thus $g(x)$ and $h(x)$ both are divisible by $x - 1$.
 Hence (C) is the correct answer.

Q.4 Find z such that $|z - 2 + 2i| \leq 1$ and z has

- (i) least absolute value (ii) numerically least amplitude.

Sol.

(i) Given equation represents a circle with centre $(2, -2)$ and radius 1.

Distance of $|z - 2 + 2i| = 1$ will be minimum from origin along the line which is normal to the circle passing through the origin, so it will passing through the centre of the circle also. $OP = (2\sqrt{2} - 1)$
 $OA = OP \cos 45^\circ$ and $OB = OP \sin 45^\circ$ so that

$$z = \frac{(2\sqrt{2} - 1)}{\sqrt{2}} - i \left(\frac{2\sqrt{2} - 1}{\sqrt{2}} \right)$$

(ii) We have to find the complex number z represent by Q where OQ is a tangent to the circle

$$\text{Now } OQ = \sqrt{OC^2 - CQ^2} = |2 - 2i|^2 = 2\sqrt{2^2 + (-2)^2 - 1} = \sqrt{7}$$

$$\therefore |z| = \sqrt{7}$$

$$\angle QOX = |\angle COX - \angle COQ|$$

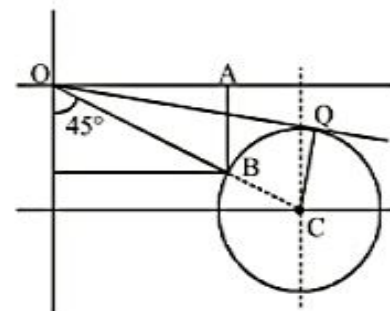
$$= - \left| \frac{\pi}{4} - \sin^{-1} \frac{1}{|2 - 2i|} \right| = - \left(\frac{\pi}{4} - \sin^{-1} \frac{1}{2\sqrt{2}} \right)$$

$$\Rightarrow \text{amp } z = - \left(\frac{\pi}{4} - \sin^{-1} \frac{1}{2\sqrt{2}} \right)$$

$$\text{amp } z = |z| \{ \cos(\text{amp } z) + i \sin(\text{amp } z) \}$$

$$= \text{amp } z = \left| \cos \left\{ - \left(\frac{\pi}{4} - \sin^{-1} \frac{1}{2\sqrt{2}} \right) \right\} + i \sin \left\{ - \left(\frac{\pi}{4} - \sin^{-1} \frac{1}{2\sqrt{2}} \right) \right\} \right|$$

$$= \text{amp } z = \left| \cos \left(\frac{\pi}{4} - \sin^{-1} \frac{1}{2\sqrt{2}} \right) - i \sin \left(\frac{\pi}{4} - \sin^{-1} \frac{1}{2\sqrt{2}} \right) \right|$$



Q.5 Plot the region represented by $\frac{\pi}{3} \leq \arg \left(\frac{z+1}{z-1} \right) \leq \frac{2\pi}{3}$ in the Argand plane.

Sol. Let us take $\arg \left(\frac{z+1}{z-1} \right) = \frac{2\pi}{3}$,

Clearly z lies on the minor arc of the circle passing through $(1, 0)$ and $(-1, 0)$.

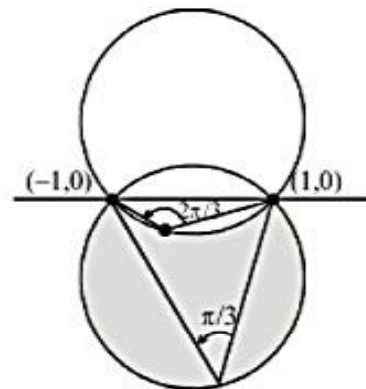
Similarly, $\arg \left(\frac{z+1}{z-1} \right) = \frac{\pi}{3}$ means that ' z ' is lying on

the major arc of the circle passing through $(1, 0)$ and $(-1, 0)$.

Now if we take any point in the region included between the two arcs, say $P_1(z_1)$,

we get $\frac{\pi}{3} \leq \arg \left(\frac{z+1}{z-1} \right) \leq \frac{2\pi}{3}$.

Thus $\frac{\pi}{3} \leq \arg \left(\frac{z+1}{z-1} \right) \leq \frac{2\pi}{3}$ represents the shaded region (excluding the points $(1, 0)$ and $(-1, 0)$).



Q.6 Consider the complex numbers $z_1 = 10 + 6i$ and $z_2 = 4 + 2i$. If $\arg \left(\frac{z - z_1}{z - z_2} \right) = \frac{\pi}{4}$, find the centre and

radius of the circle traced by the complex number z .

Sol. Let $O(z_0)$ be the centre of the circle.

We have $\angle AOB = \frac{\pi}{2}$. $AB = |z_1 - z_2| = |6 + 4i| = \sqrt{52}$.

Let $OA = OB = r \Rightarrow AB = r\sqrt{2} \Rightarrow r = \sqrt{26}$

Also $\frac{z_2 - z_0}{z_1 - z_0} = e^{-i\pi/2} = -i \Rightarrow z_2 - z_0 = -i(z_1 - z_0)$

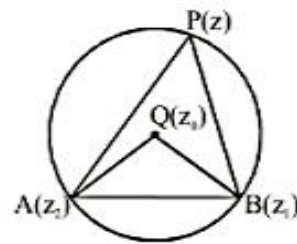
$\Rightarrow z_0 = \frac{1}{2} (z_2 - iz_2 + iz_1 + z_1) = 5 + 7i$.

Alternative:

We have $\frac{\pi}{4} = \arg \frac{z - z_1}{z - z_2} = \arg (z - z_1) - \arg (z - z_2)$

$\Rightarrow \tan^{-1} 1 = \tan^{-1} \frac{y-6}{x-10} - \tan^{-1} \frac{y-2}{x-4}$

$$\Rightarrow \tan^{-1} 1 = \tan^{-1} \left[\frac{\frac{y-6}{x-10} - \frac{y-2}{x-4}}{1 + \frac{(y-6)(y-2)}{(x-10)(x-4)}} \right]$$



$$\Rightarrow 1 = \frac{6y - 4x + 4}{x^2 + y^2 - 14x - 8y + 52}$$

$$\Rightarrow x^2 + y^2 - 10x - 14y + 48 = 0$$

$$\text{or } (x-5)^2 + (y-7)^2 = 26 = (\sqrt{26})^2$$

$$\text{or } |x-5+i(y-7)| = \sqrt{26} \Rightarrow |z-5-7i| = \sqrt{26}.$$

Q.7 Find all complex numbers z for which $\arg \left(\frac{3z-6-3i}{2z-8-6i} \right) = \frac{\pi}{4}$ and $|z-3+i| = 3$.

Sol. Now
$$\frac{3z-6-3i}{2z-8-6i} = \frac{3(x+iy)-6-3i}{2(x+iy)-8-6i}$$

$$= \frac{3x-6+i(3y-3)}{(2x-8)+i(2y-6)} \cdot \frac{[(2x-8)-i(2y-6)]}{[(2x-8)-i(2y-6)]}$$

$$= \frac{6x^2+6y^2-36x-24y+66}{(2x-8)^2+(2y-6)^2} + \frac{(12x-12y-12)}{(2x-8)^2+(2y-6)^2} = a+ib \text{ (say)}$$

Since $\arg(a+ib) = \frac{\pi}{4} \therefore \tan \frac{\pi}{4} = \frac{b}{a} \Rightarrow a=b$

$$\Rightarrow 6x^2+6y^2-36x-24y+66 = 12x-12y-12$$

$$\Rightarrow x^2+y^2-8x-2y+13=0 \quad \dots(i)$$

Again $|z-3+i|=3 \Rightarrow |x+iy-3+i|=3$

$$\Rightarrow (x-3)^2+(y+1)^2=9 \Rightarrow x^2+y^2-6x+2y+1=0 \quad \dots(ii)$$

$$(i)-(ii), \Rightarrow -2x-4y+12=0$$

$$\Rightarrow x=-2y+6 \quad \dots(iii)$$

Putting the value of x in (ii), we get

$$(-2y+6)^2+y^2-6(-2y+6)+2y+1=0$$

$$\Rightarrow 5y^2-10y+1=0$$

$$\therefore y = -\frac{10 \pm 4\sqrt{5}}{10} = 1 \pm \frac{2}{\sqrt{5}} \quad \therefore x = -2y+6 = 4 \mp \frac{4}{\sqrt{5}}$$

$$\therefore z = x+iy = 4 \mp \frac{4}{\sqrt{5}} + i \left(1 \pm \frac{2}{\sqrt{5}} \right)$$

Q.8 If $|a_n| < 2$ for $n = 1, 2, 3, \dots$ and $1 + a_1z + a_2z^2 + \dots + a_Nz^N = 0$, show that z does not lie in the interior of the circle $|z| = \frac{1}{3}$

Sol. Given $|a_n| < 2$ for $n = 1, 2, 3, \dots$
Again given $1 + a_1z + a_2z^2 + \dots + a_Nz^N = 0$
 $\Rightarrow -1 = a_1z + a_2z^2 + \dots + a_Nz^N$
 $\Rightarrow |-1| = |a_1z + a_2z^2 + \dots + a_Nz^N| = |a_1| |z| + |a_2z^2| + \dots + |a_N| |z|^N$
 $= |a_1| |z| + |a_2| |z|^2 + \dots + |a_N| |z|^N < 2|z| + 2|z|^2 + \dots + 2|z|^N$ [from (i)]
 $< 2|z| + 2|z|^2 + \dots + \text{to } \infty$ [From (iii), $|z| \neq 0$]

Case-I : When $|z| < 1$.

$$\text{From (iii), } \frac{2|z|}{1-|z|} \Rightarrow 1 - |z| < 2|z| \Rightarrow 3|z| > 1 \Rightarrow |z| > \frac{1}{3}$$

Case-II : When $|z| \geq 1$. In this case $|z| > \frac{1}{3}$ [$\because |z| \geq 1$]

Hence z does not lie in the interior of the circle $|z| = \frac{1}{3}$.

Note : Above question can also be asked as : if $|a_n| < 2$ for $1 \leq n \leq N$, then prove that there exists no

z in the interior of the circle $|z| = \frac{1}{3}$ such that

$$1 + a_1z + a_2z^2 + \dots + a_Nz^N = 0.$$

Q.9 Consider a triangle formed by the points $A\left(\frac{2}{\sqrt{3}} e^{i\frac{\pi}{2}}\right)$, $B\left(\frac{2}{\sqrt{3}} e^{-i\frac{\pi}{6}}\right)$, $C\left(\frac{2}{\sqrt{3}} e^{-i\frac{5\pi}{6}}\right)$. Let $P(z)$ be any point on its in-circle. Prove that $AP^2 + BP^2 + CP^2 = 5$.

Sol. Let $z_1 = \frac{2}{\sqrt{3}} e^{i\frac{\pi}{2}}$, $z_2 = \frac{2}{\sqrt{3}} e^{-i\frac{\pi}{6}}$, $z_3 = \frac{2}{\sqrt{3}} e^{-i\frac{5\pi}{6}}$.

Clearly the points lie on the circle $|z| = \frac{2}{\sqrt{3}}$

If l the length of side of the $\triangle ABC$, $AD = l \sin 60^\circ = l \frac{\sqrt{3}}{2} \Rightarrow OD = \frac{1}{3} AD = \frac{l\sqrt{3}}{3}$

$$\text{Now, } OA = \frac{2}{3} AD = \frac{2}{3} \cdot \frac{l\sqrt{3}}{2} = \frac{l\sqrt{3}}{3}$$

$$\Rightarrow l = 2 \text{ and } OD = \frac{2\sqrt{3}}{6} = \frac{1}{\sqrt{3}}$$

Let $P(z)$ be any point on the in circle, is $|z| = \frac{1}{\sqrt{3}}$

$$\Rightarrow z = \frac{1}{\sqrt{3}} e^{i\theta}$$

$$\text{Now, } AP^2 = |z - z_1|^2 = |z|^2 + |z_1|^2 - (z\bar{z}_1 + \bar{z}z_1)$$

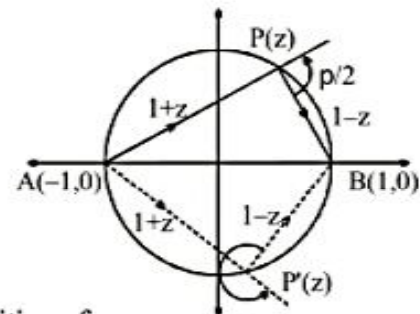
$$\text{Similarly, } BP^2 = |z|^2 + |z_2|^2 - (z\bar{z}_2 + \bar{z}z_2) \text{ and } CP^2 = |z|^2 + |z_3|^2 - (z\bar{z}_3 + \bar{z}z_3)$$

$$\begin{aligned} AP^2 + BP^2 + CP^2 &= 3|z|^2 + |z_1|^2 + |z_2|^2 + |z_3|^2 - z(\bar{z}_1 + \bar{z}_2 + \bar{z}_3) - \bar{z}(z_1 + z_2 + z_3) \\ &= 1 + 3 \cdot \frac{4}{3} - z(0) - \bar{z}(0) = 5. \end{aligned}$$

Q.10 If $|z| = 1$, then prove that the points represented by $\sqrt{\frac{1+z}{1-z}}$ lie on one or other of two fixed perpendicular straight lines.

Sol. Since $|z| = 1$, z lies on a unit circle having centre at the origin

$$\begin{aligned} \arg\left(\frac{1+z}{1-z}\right) &= +\frac{\pi}{2} \text{ or } +\frac{3\pi}{2} \\ \Rightarrow \frac{1+z}{1-z} &= ke^{i\pi/2} \text{ or } ke^{i3\pi/2} \end{aligned}$$



where k is a real parameter and its value depends upon the position of z .

$$\begin{aligned} \text{Let } \alpha &= \sqrt{\frac{1+z}{1-z}} \Rightarrow \alpha = \sqrt{k} e^{i\pi/4} \text{ or } \sqrt{k} e^{i3\pi/4} \\ \Rightarrow \alpha &\text{ lies on one or other of the two perpendicular lines.} \end{aligned}$$

Q.11 Dividing $f(z)$ by $z-i$, we obtain the remainder i and dividing it by $z+i$, we get remainder $1+i$. Find the remainder upon the division of $f(z)$ by z^2+1 .

Sol. $z-i=0 \Rightarrow z=i$

Remainder when $f(z)$ is divided by $(z-i) = f(i)$

Similarly remainder when $f(z)$ is divided by $(z+i) = f(-i)$... (i)

According to question $f(i) = i$

and $f(-i) = 1+i$... (ii)

Since z^2+1 is a quadratic expression, therefore remainder when $f(z)$ is divided by z^2+1 will be in general a linear expression.

Let $g(z)$ be the quotient and $az+b$ the remainder when $f(z)$ is divided by z^2+1 .

$$\text{then } f(z) = g(z)(z^2+1) + az+b \quad \dots \text{(iii)}$$

$$\therefore f(i) = g(i)(i^2+1) + ai+b = ai+b \quad \dots \text{(iv)}$$

$$\text{and } f(-i) = g(-i)(i^2+1) - ai+b = -ai+b \quad \dots \text{(v)}$$

$$\text{From (i) and (iv), we have } b+ai = i \quad \dots \text{(vi)}$$

$$\text{From (ii) and (v), we have } b-ai = 1+i \quad \dots \text{(vii)}$$

$$\text{Solving (vi) and (vii) we get, } b = \frac{1}{2} + i \text{ and } a = \frac{i}{2}$$

$$\text{Hence required remainder } = az+b = \frac{1}{2}iz + \frac{1}{2} + i$$

Paragraph for question nos. 12 to 14

Let A, B, C be three sets of complex numbers as defined below.

$$A = \{z : |z+1| \leq 2 + \operatorname{Re}(z)\}, \quad B = \{z : |z-1| \geq 1\} \quad \text{and} \quad C = \left\{z : \left| \frac{z-1}{z+1} \right| \geq 1\right\}$$

Q.12 The number of point(s) having integral coordinates in the region $A \cap B \cap C$ is
(A) 4 (B) 5 (C) 6 (D) 10

Q.13 The area of region bounded by $A \cap B \cap C$ is
(A) $2\sqrt{3}$ (B) $\sqrt{3}$ (C) $4\sqrt{3}$ (D) 2

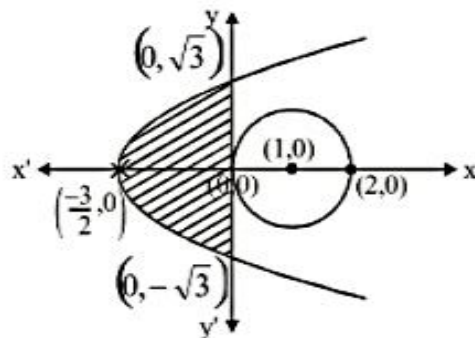
Q.14 The real part of the complex number in the region $A \cap B \cap C$ and having maximum amplitude is
(A) -1 (B) $-\frac{3}{2}$ (C) $\frac{1}{2}$ (D) -2

Sol. For A, $|z+1| \leq 2 + \operatorname{Re}(z)$
 $\Rightarrow (x+1)^2 + y^2 \leq 4 + 4x + x^2$
 $\Rightarrow y^2 \leq 3 + 2x$

$$\Rightarrow y^2 \leq 2\left(x + \frac{3}{2}\right) \quad \dots(1)$$

For B, $|z-1| \geq 1$
 $\Rightarrow (x-1)^2 + y^2 \geq 1 \quad \dots(2)$

For C, $|z-1|^2 \geq |z+1|^2$
 $\Rightarrow (z-1)(\bar{z}-1) \geq (z+1)(\bar{z}+1)$
 $\Rightarrow (z\bar{z} - \bar{z} - z + 1) \geq (z\bar{z} + \bar{z} + z + 1)$
 $\Rightarrow z + \bar{z} \leq 0$
 i.e. $x \leq 0 \quad \dots(3)$



(i) $(-1,0), (-1,1), (-1,-1), (0,0), (0,1), (0,-1)$ but $z = -1$ is not in the domain in set C
 $\therefore$ Total number of point(s) having integral coordinates in the region $A \cap B \cap C$ is 5.

(ii) Required area $= 2 \int_{-\frac{3}{2}}^0 \sqrt{2\left(x + \frac{3}{2}\right)} dx = 2\sqrt{3}$ (square units)

(iii) Clearly $z = -\frac{3}{2} + i0$ is the complex number in the region $A \cap B \cap C$ and having maximum amplitude.

$$\therefore \operatorname{Re}(z) = -\frac{3}{2}$$

Paragraph for question nos. 15 to 17

Let $A(z_1)$ be the point of intersection of curves $\arg(z - 2 + i) = \frac{3\pi}{4}$ and $\arg(z + i\sqrt{3}) = \frac{\pi}{3}$.

$B(z_2)$ be the point on the curve $\arg(z + i\sqrt{3}) = \frac{\pi}{3}$ such that $|z_2 - 5|$ is minimum and $C(z_3)$ be the centre of circle $|z - 5| = 3$.

[Note : $i^2 = -1$]

Q.15 The area of triangle ABC is equal to

- (A) $4\sqrt{3}$ (B) $\frac{3\sqrt{3}}{2}$ (C) $2\sqrt{3}$ (D) 4

Q.16 The equation of straight line passing through origin and perpendicular to line joining $A(z_1)$ and $B(z_2)$ on the complex plane is equal to

- (A) $z = \lambda(2 + i\sqrt{3})$ (B) $z = \lambda(-\sqrt{3} + i)$ (C) $z = \lambda(1 + i\sqrt{3})$ (D) $z = \lambda(\sqrt{3} + i)$
(where λ is real parameter.)

Q.17 If $|z - z_1| = 1$ and $\omega = \operatorname{Re}(z + 2)$ then ω lie on

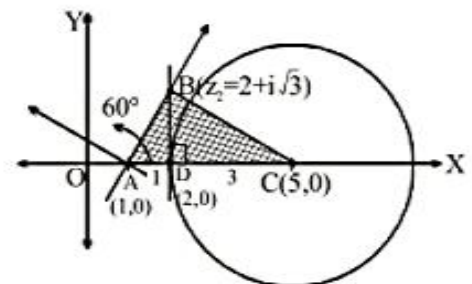
- (A) real axis. (B) line not passing through the origin.
(C) line segment joining (2, 0) and (4, 0). (D) circle centred at (1, 0) and radius 2.

Sol.(i) Clearly, $A(z_1)$ is the point of intersection of $\arg(z - 2 + i) = \frac{3\pi}{4}$ and $\arg(z + i\sqrt{3}) = \frac{\pi}{3}$

$$\Rightarrow z_1 = 1$$

Also, $B(z_2)$ is the point on $\arg(z + i\sqrt{3}) = \frac{\pi}{3}$

such that $|z_2 - 5| = \text{minimum}$, so $z_2 = 2 + i\sqrt{3}$.



Ans $C(z_3)$ be the centre of the circle $|z - 5| = 3 \Rightarrow z_3 = 5$.

$$\text{Hence, area } (\Delta ABC) = \frac{1}{2} (AC) (AD) = \frac{1}{2} (4) (\sqrt{3}) = 2(\sqrt{3}) \text{ (square units.)}$$

(ii) Clearly, required equation of straight line is

$$z = \lambda i (z_2 - z_1) = \lambda i (2 + i\sqrt{3} - 1) = \lambda i (1 + i\sqrt{3}) = \lambda (-\sqrt{3} + i).$$

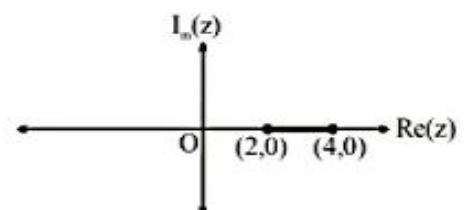
(iii) We have

$$|z - 1| = 1 \Rightarrow z = 1 + e^{i\theta}, \theta \in [0, 2\pi)$$

$$\Rightarrow z = 1 + e^{i\theta} = 1 + \cos \theta + i \sin \theta$$

$$\therefore \omega = \operatorname{Re}(z + 2) = 3 + \cos \theta = \text{purely real}$$

Clearly, locus of ω is the line segment joining (2, 0) and (4, 0). Ans.



Q.18 z_1 & z_2 are two distinct points in an argand plane. If $a|z_1| = b|z_2|$, (where $a, b \in \mathbb{R}$) then prove that the point $\frac{az_1}{bz_2} + \frac{bz_2}{az_1}$ lies on the line segment $[-2, 2]$ of the real axis.

Sol. Assuming $\arg z_1 = \theta$ and $\arg z_2 = \theta + \alpha$.

$$\frac{az_1}{bz_2} + \frac{bz_2}{az_1} = \frac{a|z_1|e^{i\theta}}{b|z_2|e^{i(\theta+\alpha)}} + \frac{b|z_2|e^{i(\theta+\alpha)}}{a|z_1|e^{i\theta}} = e^{-i\alpha} + e^{i\alpha} = 2 \cos \alpha$$

Alternatively: Let $\alpha = \frac{az_1}{bz_2}$; $\frac{1}{\alpha} = \frac{bz_2}{az_1}$; Also $|\alpha| = \frac{|az_1|}{|bz_2|} = \frac{a|z_1|}{b|z_2|} = 1 \Rightarrow \alpha = \frac{1}{\alpha}$

$$\Rightarrow \alpha + \frac{1}{\alpha} = \alpha + \bar{\alpha} = 2 \operatorname{Re}(\alpha) = 2 \cos \alpha$$

Q.19 If z_1, z_2, z_3 are complex numbers such that $\frac{2}{z_1} = \frac{1}{z_2} + \frac{1}{z_3}$, show that the points represented by z_1, z_2, z_3 lie on a circle passing through the origin.

Sol. Since $P(z_1), Q(z_2), R(z_3)$ and $S(z_4)$ are concyclic points,

$$\angle PSQ = \angle PRQ \Rightarrow \arg \frac{z_2 - z_4}{z_1 - z_4} = \arg \frac{z_2 - z_3}{z_1 - z_3}$$

$$\Rightarrow \arg \left[\left(\frac{z_2 - z_4}{z_1 - z_4} \right) \left(\frac{z_1 - z_3}{z_2 - z_3} \right) \right] = 0$$

$$\Rightarrow \frac{(z_2 - z_4)}{(z_1 - z_4)} \cdot \frac{(z_1 - z_3)}{(z_2 - z_3)} = \text{real}$$

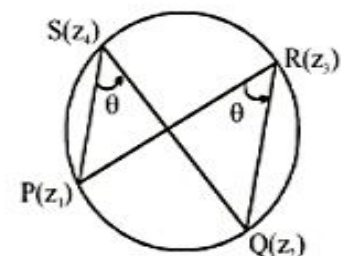
If $z_4 = 0 + i0$, then $\frac{z_2}{z_1} \cdot \frac{z_1 - z_3}{z_2 - z_3} = \text{real} \quad \dots\dots(1)$

We have $\frac{2}{z_1} = \frac{1}{z_2} + \frac{1}{z_3}$ from which $z_3 = \frac{z_1 z_2}{2z_2 - z_1} \quad \dots\dots(2)$

From (1) and (2), $\frac{z_2}{z_1} \times \frac{z_1 - \frac{z_1 z_2}{2z_2 - z_1}}{z_2 - \frac{z_1 z_2}{2z_2 - z_1}} = \text{real}$

$$\Rightarrow \frac{z_2 - z_1}{2(z_2 - z_1)} = \text{real} \Rightarrow \frac{1}{2} = \text{real, which is true.}$$

Therefore, z_1, z_2, z_3 and the origin are concyclic.



Alternative

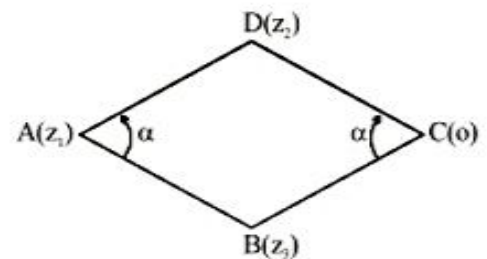
$$\frac{2}{z_1} = \frac{1}{z_2} + \frac{1}{z_3} \Rightarrow \frac{1}{z_1} - \frac{1}{z_2} = \frac{1}{z_3} - \frac{1}{z_1}$$

$$\Rightarrow \frac{z_2 - z_1}{z_1 z_2} = \frac{z_1 - z_3}{z_1 z_3} \Rightarrow \frac{z_2 - z_1}{z_3 - z_1} = -\frac{z_2}{z_3}$$

$$\Rightarrow \arg \left(\frac{z_2 - z_1}{z_3 - z_1} \right) = \arg \left(-\frac{z_2}{z_3} \right) = \pi + \arg \left(\frac{z_2}{z_3} \right)$$

$$\Rightarrow \alpha = \pi - \beta \Rightarrow \alpha + \beta = \pi$$

$\Rightarrow$ points A, B, C, D are concyclic.

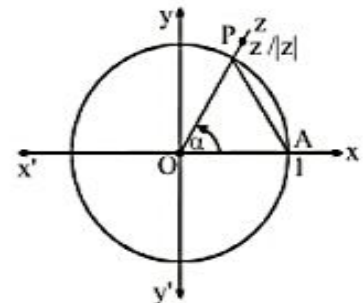


Q.20 Prove the following inequalities geometrically and analytically:

$$(a) \quad \left| \frac{z}{|z|} - 1 \right| \leq |\arg z| \quad (b) \quad |z - 1| \leq |z| + |\arg z|$$

Sol.(a) The quantity $\frac{z}{|z|}$ lies on a unit circle centered at the origin.

It is clear that $\left| \frac{z}{|z|} - 1 \right| = AP \leq \text{arc}(AP) = \alpha = \arg(z)$



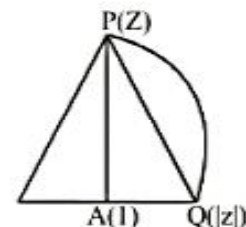
If α is negative, this gives $\left| \frac{z}{|z|} - 1 \right| \leq |\arg z|$.

To show this analytically, put $z = r(\cos \alpha + i \sin \alpha)$,

where $r = |z|$. Then $\left| \frac{z}{|z|} - 1 \right| = |(\cos \alpha - 1) + i \sin \alpha|$

$$= [(\cos \alpha - 1)^2 + \sin^2 \alpha]^{1/2} = [2(1 - \cos \alpha)]^{1/2}$$

$$= \left(4 \sin^2 \frac{\alpha}{2} \right)^{1/2} = 2 \left| \sin \frac{\alpha}{2} \right| \leq 2 \left| \frac{\alpha}{2} \right| = |\alpha| = |\arg z| \quad [\sin \theta \leq \theta \text{ if } \theta \geq 0]$$



(b) Referring to the show figure and usign the result of part (a),

$$\text{we have } |z-1| = AP \leq AQ + QP = ||z|-1| + |z-z|$$

$$\Rightarrow |z-1| \leq ||z|-1| + |z| \left| \frac{z}{|z|} - 1 \right| \leq ||z|-1| + |z| |\arg z|.$$

Analytically, we can get this result as follows :

$$|z-1| = |(z-|z|) + (|z|-1)| \leq |z-|z|| + ||z|-1|$$

$$\Rightarrow |z-1| \leq ||z|-1| + |z| \left| \frac{z}{|z|} - 1 \right| \leq ||z|-1| + |z| |\arg z|.$$

Q.21 If $z = 2 + t + i\sqrt{3-t^2}$, where t is real and $t^2 < 3$, show that the modulus of $\frac{z+1}{z-1}$ is independent of

t . Also show that the locus of the points z for different values of t is a circle and find its centre and radius.

Sol. We have $\frac{z+1}{z-1} = \frac{3+i+i\sqrt{3-t^2}}{1+t+i\sqrt{3-t^2}}$

$$\therefore \left| \frac{z+1}{z-1} \right|^2 = \frac{(3+t)^2 + 3-t^2}{(1+t)^2 + 3-t^2} = \frac{6(t+2)}{2(t+2)} = 3 \quad (\text{independent of } t)$$

Let $z = x + iy$, then $x + iy = 2 + t + i\sqrt{3-t^2}$

Equating real and imaginary parts, we get

$$x = 2 + t \quad \dots(i) \quad \text{and} \quad y = \sqrt{3-t^2} \quad \dots(ii)$$

[In order to find the locus of z we will have a eliminate t]

Putting the value of t from (i) and (ii), we get

$$y^2 = 3 - (x-2)^2 \quad \text{or} \quad (x-2)^2 + y^2 = 3$$

Thus locus of z is a circle whose centre is $(2, 0)$ and radius is $\sqrt{3}$.

Q.22 $|z| \leq 1, |w| \leq 1$, show that $|z - w|^2 \leq (|z| - |w|)^2 + (\arg z - \arg w)^2$.

Sol. Let us consider a unit circle with its centre as the origin.

Let $\angle AOX = \theta_1$ and $\angle BOX = \theta_2$

$\therefore \arg(z) = \theta_1$ and $\arg(w) = \theta_2$.

$z = \overline{OA}, w = \overline{OB}$

Now in $\triangle OAB$,

$$\cos \theta = \frac{|\overline{OA}|^2 + |\overline{OB}|^2 - |\overline{AB}|^2}{2|\overline{OA}||\overline{OB}|}$$

$$\Rightarrow |\overline{AB}|^2 = |\overline{OA}|^2 + |\overline{OB}|^2 - 2|\overline{OA}||\overline{OB}|\cos \theta$$

$$\Rightarrow |z - w|^2 = |z|^2 + |w|^2 - 2|z||w|\cos \theta = (|z| - |w|)^2 - 4|z||w|\sin^2\left(\frac{\theta}{2}\right)$$

$$\text{We know } \sin \frac{\theta}{2} \leq \frac{\theta}{2} \quad \left[0 \leq \frac{\theta}{2} < \frac{\pi}{2}\right]$$

$$\text{Hence } |z - w|^2 \leq (|z| - |w|)^2 + (\arg z - \arg w)^2$$

